

RESEARCH THESIS

Decision Intelligence System for Satellite Constellation Optimization under Space Weather Uncertainty

*A Unified Forecasting, Risk-Scoring, and Optimization Framework Integrating Deep Learning,
Orbital Mechanics, Stochastic Simulation, and Integer Programming*

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Abstract

Satellites that provide communication, navigation, and Earth-observation services share a common, largely invisible adversary: the Sun. Solar flares, coronal mass ejections, and the resulting geomagnetic storms inject energetic particles into near-Earth space, degrading radio links, corrupting onboard electronics, and threatening the very orbits that the global space economy depends on. Despite this, the great majority of operational satellite scheduling systems treat space weather as background noise rather than as a first-class input to be forecast, quantified, and acted upon.

This thesis presents a complete, end-to-end Decision Intelligence System that closes this gap. The system ingests multi-channel space weather telemetry (electron flux, solar wind speed, interplanetary magnetic field, soft and hard X-ray flux), forecasts near-term radiation conditions using storm-aware Long Short-Term Memory (LSTM) networks with attention and Monte Carlo dropout uncertainty bands, and propagates that uncertainty all the way through to a fourteen-satellite constellation spanning GEO, IGSO, LEO, MEO, and Sun-synchronous orbits. Each satellite receives a continuously updated, explainable risk score derived from radiation exposure, flare probability, geomagnetic storm intensity, orbital geometry, and solar illumination angle. A four-state Markov chain models long-run operational reliability (Active, Communicating, Idle, Faulty); Markov Chain Monte Carlo (MCMC) sampling via the Metropolis-Hastings algorithm quantifies orbital position uncertainty; and an Integer Linear Program (ILP), solved with a branch-and-bound CBC solver, computes the provably optimal assignment of satellites to ground stations subject to visibility, antenna-capacity, and service-level constraints.

A companion thesis, developed alongside this system, extends the scheduling layer into a Hybrid Satellite-Ground Communication Architecture: a rigorously modeled proposal showing that a dense, fiber-interconnected ground-station network can substitute for orbital satellite mass, easing the collision and debris pressures of mega-constellation growth. Both systems share the same mathematical backbone — orbital mechanics, visibility geometry, MCMC uncertainty, Markov reliability, and ILP optimization — and are presented together here as a single, coherent body of work.

The system was validated twice: once against a fully synthetic 4,320-sample dataset engineered to exercise flare and storm dynamics, and once against 44,640 minutes of real geomagnetic index data (K_p , A_p , $F_{10.7}$) from the GFZ Potsdam observatory covering the May 2024 Gannon superstorm — the most intense geomagnetic event in two decades. All twelve physics-based ground-truth assertions passed against the real event, and the synthetic and real datasets were shown to agree in structure and timing while differing, as expected, in fine-grained noise character. An honest accounting of forecast skill is provided throughout: an initial naive LSTM configuration produced negative skill scores during storms (worse than a persistence baseline), and the diagnosis and correction of that failure — through storm-aware sample weighting, an isolated target-scaling pipeline, and a removed clipping bug — is documented as a central methodological contribution, not hidden as an embarrassment.

The result is a working decision-intelligence pipeline that converts raw space-weather telemetry into concrete, explainable operator commands — reduce load, activate backup path, enter safe mode — within seconds, while simultaneously producing a constellation-wide optimization that respects physical and operational constraints. Both the underlying Google Colab pipeline (`radiation_solar_flares.py`, Cells 1–17) and the companion CesiumJS/Three.js 3D visualization were executed and validated, and representative captures of both running systems are reproduced throughout this thesis to document exactly what each module produces and how an operator interacts with it. The thesis documents this system at three levels throughout: a plain-language explanation for a general reader, a rigorous technical and mathematical treatment for a domain specialist, and a real-world analogy to anchor

intuition, so that the work is legible to mission operators, students, and machine learning researchers alike.

Keywords: *Space Weather Forecasting, Satellite Constellation Scheduling, LSTM with Attention, Monte Carlo Dropout, Markov Chain Reliability, Metropolis-Hastings MCMC, Integer Linear Programming, Decision Intelligence, Orbital Mechanics, Gannon Storm Validation.*

1. Introduction

1.1 Problem Statement

Modern satellite operations face a persistent and growing risk that originates almost 150 million kilometers away, on the surface of the Sun. Solar flares and coronal mass ejections (CMEs) accelerate streams of charged particles toward Earth. When these particles arrive, they raise the radiation environment in near-Earth orbits, increase atmospheric drag, disturb the geomagnetic field, and directly degrade the radio links and electronics that satellites depend on. The Indian Space Research Organisation (ISRO) frames this challenge through two linked problem statements that this thesis addresses jointly: Problem Statement 14, radiation forecasting for satellite electronics, and Problem Statement 15, solar flare prediction for operational alerting.

The central engineering question this thesis answers is: given continuously streaming, noisy, and partially unreliable space-weather telemetry, how should a constellation of satellites in different orbits be monitored, scored for risk, and re-scheduled in near real time, in a way that is both mathematically optimal and operationally explainable to a human controller?

IN PLAIN TERMS

Think of a national power grid during an incoming hurricane. The grid operator does not wait for transformers to start failing. They watch the storm's forecast track, identify which substations are most exposed, pre-emptively shed load on the riskiest ones, and reroute power through safer lines — all before the storm makes landfall. This thesis builds the equivalent system for satellites and solar storms: forecast the storm, score each satellite's exposure, and reroute communication load before anything actually breaks.

1.2 Motivation: Space Weather Risks and Satellite Failures

The motivation for this work is not theoretical. Energetic electrons trapped in the Van Allen belts can accumulate as deep-dielectric charge inside spacecraft components, eventually discharging and causing single-event upsets, memory corruption, or permanent component failure. Solar X-ray bursts ionize the upper atmosphere within minutes, disrupting high-frequency radio communication on the sunlit side of Earth almost instantaneously. Geomagnetic storms, measured by the Disturbance Storm Time (Dst) index, increase atmospheric density at low orbital altitudes, accelerating orbital decay and complicating station-keeping.

The May 2024 Gannon storm, used as a real-world validation case throughout this thesis, illustrates the scale of the problem concretely. During this event the planetary Kp index reached its maximum possible value of 9.0, the Ap index peaked at 400, solar wind speed exceeded 1,000 km/s, and the interplanetary magnetic field's north-south component (Bz) swung to -55 nT — a deeply southward orientation that couples efficiently with Earth's magnetosphere. Operators of multiple satellite fleets, including commercial LEO constellations, reported measurable orbital decay and required active maneuvering in the days following the storm. A decision-support system that can see this kind of event coming, even by minutes to hours, and translate it into specific per-satellite actions has direct, quantifiable value.

1.3 Objectives

This thesis pursues five concrete, interlocking objectives:

1. Forecast near-term space weather conditions — solar flare probability and multi-horizon electron flux — using sequence models trained on physically structured features, with honest, non-inflated accuracy reporting.
2. Model satellite orbital geometry precisely enough to compute, for any future time, which ground stations can physically communicate with which satellites, using Newtonian orbital mechanics and Earth-Centered Inertial (ECI) coordinate transforms.
3. Quantify, rather than ignore, the uncertainty in both space weather forecasts and orbital positions, using Monte Carlo dropout for the former and Metropolis-Hastings MCMC sampling for the latter.
4. Translate continuous risk into discrete, explainable operational classifications and actions for every satellite in a heterogeneous fourteen-satellite fleet (six ISRO assets plus an eight-satellite constellation spanning four orbital families), with a transparent 'why' behind every decision.
5. Optimize the assignment of satellites to ground stations under realistic operational constraints — line-of-sight visibility, antenna capacity, and service-level guarantees — using Integer Linear Programming with a certified-optimal solver, and to do so robustly under positional uncertainty via a Stochastic Integer Program.

WHY THIS APPROACH

Each objective targets a distinct failure mode of naive approaches: objective 1 fails if you ignore physics-informed features; objective 2 fails if you assume satellites are always reachable; objective 3 fails if you report a single number and call it certain; objective 4 fails if your system outputs a probability with no explanation a controller can act on; and objective 5 fails if you accept a 'good enough' heuristic schedule when a certifiably optimal one is computable in seconds.

2. Literature Review

2.1 Satellite Constellation Optimization

The scheduling of satellite-to-ground contacts has classically been formulated as a combinatorial optimization problem closely related to job-shop scheduling and interval scheduling. Once a discrete time grid and a binary visibility matrix are defined, the contact-scheduling problem becomes an assignment problem with capacity and exclusivity constraints — a structure well suited to Integer Linear Programming (ILP). Nemhauser and Wolsey's foundational treatment of integer and combinatorial optimization, and Luenberger and Ye's treatment of linear and nonlinear programming, provide the mathematical machinery (branch-and-bound, LP relaxation, duality) underlying the CBC solver used throughout this work.

The companion Hybrid Satellite-Ground Communication Architecture thesis extends this literature in a specific direction: rather than treating the number of satellites as fixed and optimizing only the schedule, it treats ground-station density as a second decision variable, exploring the economic and combinatorial trade-off between orbital and terrestrial infrastructure. This connects the scheduling literature to the growing space-traffic-management literature on mega-constellation congestion (Kessler & Cour-Palais's original 1978 analysis of collision cascades remains the seminal reference point for this risk).

2.2 Space Weather Forecasting

Operational space-weather forecasting has historically relied on empirical relationships between geomagnetic indices (Kp, Ap, Dst) and particle flux, such as the Zhelavskaya et al. (2019) electron-flux relationship used in this thesis's real-data validation pipeline, and on physics-based magnetospheric models. The Burton et al. (1975) Dst injection equation, which this thesis inverts to derive a synthetic interplanetary magnetic field proxy, remains a standard reference for relating solar wind energy input to geomagnetic disturbance.

More recently, the field has absorbed sequence-modeling techniques from machine learning. Recurrent architectures, and LSTM networks in particular, are well suited to space weather because the underlying physical processes — flare onset, CME propagation, magnetospheric ring current decay — are temporally structured with both fast transients (flares, over minutes) and slow trends (solar rotation, over 27 days). A persistent challenge documented in the forecasting literature, and confronted directly in Section 7 of this thesis, is class imbalance: storms are rare events in any reasonably long training window, and naive loss functions trained on mostly-quiet data systematically under-predict the rare, high-magnitude spikes that matter most operationally.

2.3 Artificial Intelligence in Aerospace

AI methods have entered aerospace operations along several fronts: anomaly detection in telemetry streams, autonomous fault diagnosis, trajectory optimization, and increasingly, end-to-end decision support that sits above individual subsystem models. This thesis's contribution sits in the last category. Rather than treating the LSTM forecaster, the Markov reliability model, and the ILP scheduler as independent academic exercises, it integrates them into a single pipeline where the output of one model becomes a structured input to the next — a forecast's upper confidence bound feeds a risk score;

a risk score feeds a classification; a classification feeds an action; and the underlying reliability estimate feeds the optimization's feasibility constraints.

2.4 Decision Intelligence Systems

Decision intelligence, as a discipline, is distinguished from pure predictive modeling by its explicit closure of the loop from data to forecast to uncertainty to recommended action, typically with an explainability layer that lets a human operator audit and override the system. This thesis adopts that discipline deliberately: every risk score produced by the system in Section 14 is accompanied by a structured explanation block enumerating exactly which physical factors (elevated radiation, flare probability, geomagnetic storm intensity, low elevation angle, sun-facing orbit geometry) contributed to the score and by how much. This design choice reflects a recurring theme in the decision-intelligence literature: a system that a controller cannot interrogate is a system a controller will eventually stop trusting, regardless of its underlying accuracy.

WHY THIS APPROACH

The literature converges on a clear lesson that shapes this thesis's architecture: forecasting accuracy, uncertainty quantification, and explainability are not three competing priorities to trade off against each other — they are sequential layers, each of which is necessary but insufficient on its own. A highly accurate forecast with no uncertainty band is unsafe to act on; an uncertainty-aware forecast with no explainability layer is unauditable; and an explainable system built on an inaccurate forecast is confidently wrong. This thesis is structured precisely to deliver all three layers in series.

3. System Overview

3.1 End-to-End Architecture

The system is organized as a fifteen-stage pipeline, implemented across two parallel research artifacts that share a common mathematical core: a Google Colab notebook (radiation_solar_flares.py, restructured here from its original cell-by-cell Colab form, spanning Cells 1 through 17) implementing the full forecasting, risk, and decision layers; and an interactive Three.js/CesiumJS dashboard (ISRO_SpaceWeather_3D.html) rendering the same orbital and risk model as a live, explorable 3D scene. Table 3.1 summarizes every stage, its method, and its output.



Figure 3.1. The ISRO Space Weather Monitoring Dashboard, generated directly by Cells 1–2 of radiation_solar_flares.py. Soft X-ray flux, hard X-ray flux, >2 MeV electron flux, and the Dst index proxy are plotted across the 72-hour synthetic validation window, with the four catalogued flare events (Section 6.3) marked as vertical dashed lines. This is the first static output the pipeline produces and the visual anchor for the entire Forecasting Loop described in Section 3.2.

Data flows top to bottom: raw telemetry enters at Stage 1 and is progressively transformed into forecasts (Stages 3–4), physical quantities (Stages 5–9), reliability and risk (Stages 10–13), and finally an optimized, executable schedule (Stage 14), rendered for human inspection at Stage 15.

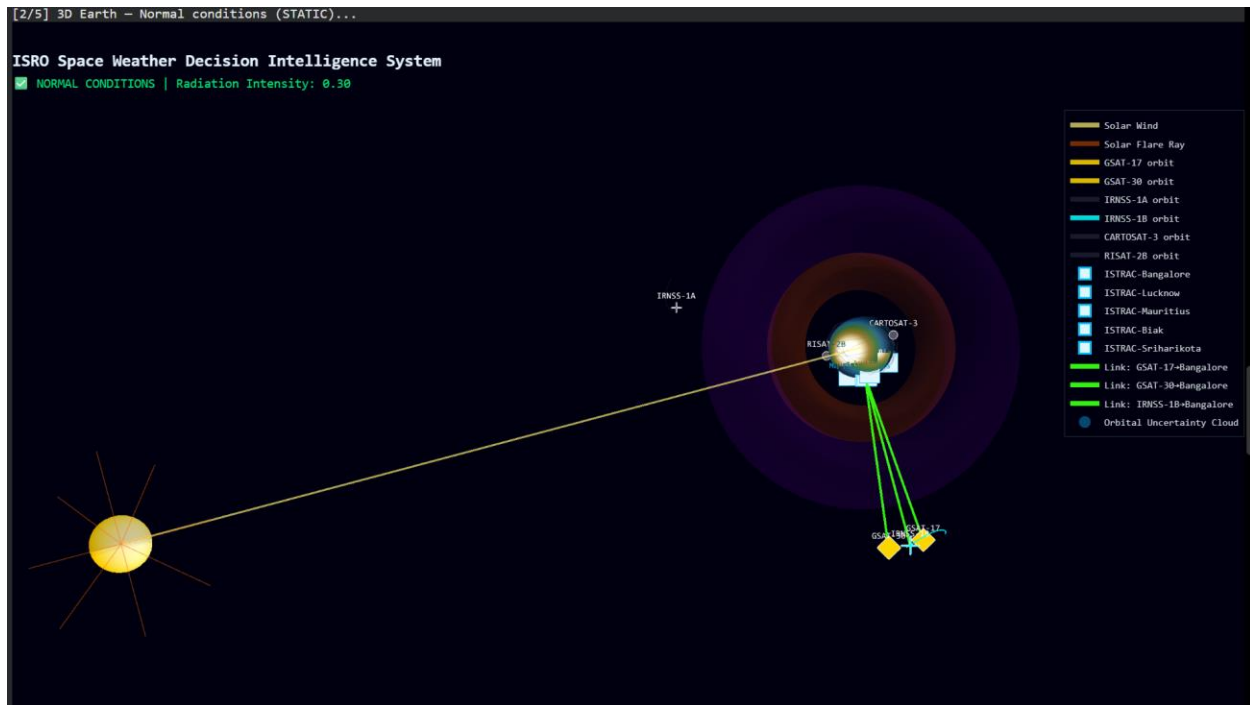


Figure 3.2. The live 3D constellation view (*ISRO_SpaceWeather_3D.html*), rendered in Three.js/WebGL and captured here under normal space-weather conditions (radiation intensity = 0.30, status banner GREEN). The scene shows the Sun, the solar-wind vector traveling toward Earth, the radiation field surrounding Earth as concentric colored rings, all six ISRO satellite orbits (GSAT-17, GSAT-30, IRNSS-1A, IRNSS-1B, CARTOSAT-3, RISAT-2B), the five ISTRAC ground-station markers, and the active green communication links — the same orbital and risk model described mathematically in Sections 9–14, rendered here for live operator inspection. Section 3.4 below explains how an operator drives this same scene interactively.

Table 3.1 — The Fifteen-Stage Pipeline

#	Stage	Method	Output
1	Data Ingestion	Multi-channel telemetry (electron flux, solar wind, IMF Bz, proton density, soft/hard X-ray)	6-channel master_df (synthetic, 4,320 samples; real, 44,640 samples)
2	Flare Nowcasting	Rolling-baseline spike detection on X-ray channels	Flare catalog with class (A–X), peak time, peak flux
3	Forecasting (PS15)	LSTM + Attention, binary cross-entropy	30-min-ahead flare probability
4	Forecasting (PS14)	Storm-aware LSTM + MC-Dropout, multi-horizon	30-min / 6-hr / 12-hr electron flux $\pm 1\sigma$
5	Physics Layer	Propagation delay, Dst proxy, radiation field synthesis	Radiation intensity [0,1], Dst index (nT)

#	Stage	Method	Output
6	Orbital Mechanics	Kepler + Newton ECI propagation, elevation geometry	Satellite ECI/ECEF trajectories, elevation angle time series
7	Visibility Model	Threshold filter on elevation angle	Binary visibility matrix $V[\text{sat, ground-station, time}]$
8	Radiation–Comm Coupling	Shannon capacity under radiation-degraded SNR	Channel capacity (Mbps), fault probability
9	Uncertainty (MCMC)	Metropolis-Hastings sampling of orbital elements	Position uncertainty cloud, σ per satellite
10	Reliability (Markov)	4-state chain, radiation-modified transition matrix	Stationary distribution, time-varying reliability
11	Risk Scoring	Weighted multi-factor score with geometric modifiers	Per-satellite risk score $[0,1]$ + explanation
12	Classification	Threshold bands on risk score	SAFE / CAUTION / HIGH_RISK / CRITICAL
13	Decision Engine	Rule-based action mapping + explainability block	Per-satellite action, new load, 'why' reasons
14	Optimization (ILP)	Binary program, CBC branch-and-bound	Optimal satellite–ground-station assignment
15	Visualization	Plotly / Three.js dashboards	3D constellation view, Gantt, heatmaps, live explorer

Table 3.1. The fifteen-stage Decision Intelligence pipeline, method employed at each stage, and the structured output passed to the next stage.

3.2 Flow Diagram Explanation

It is useful to read the architecture as three concentric loops rather than a single straight line.

1. The Forecasting Loop (Stages 1–4) ingests raw telemetry and produces probabilistic statements about the near future: a flare probability and an electron-flux forecast with an explicit uncertainty band. This loop runs purely in time — it has no knowledge of satellites yet.
2. The Physical Loop (Stages 5–9) computes where every satellite actually is, whether it can see a ground station right now, and how confident we are in that position given uncertainty in the underlying orbital elements. This loop runs purely in geometry — it has no knowledge of forecasts yet.
3. The Decision Loop (Stages 10–14) is where the two upstream loops merge. The forecast's upper confidence bound becomes an input to each satellite's risk score; the satellite's orbit type and current elevation become geometric modifiers on that same score; the resulting classification

drives a recommended action; and the full set of per-satellite reliabilities and visibilities becomes the constraint set for the final ILP optimization.

REAL-WORLD ANALOGY

This three-loop structure mirrors how a hospital emergency department actually operates, not how it is drawn on an org chart. One team continuously monitors incoming ambulance reports and triage signals (the Forecasting Loop). A second team tracks which beds, surgeons, and equipment are physically available right now (the Physical Loop). The decision of which patient goes to which resource, and in what order, only happens where those two streams of information meet — and that decision is re-made continuously as new ambulances arrive and new beds free up, exactly as this system re-solves its ILP at every time step.

3.3 Software and Computational Stack

The forecasting and decision layers are implemented in Python using PyTorch for the LSTM models, scikit-learn for scaling and evaluation metrics, NumPy and SciPy for the physics, MCMC, and Markov computations, and PuLP with the CBC backend for Integer Linear Programming. Visualization uses Plotly for the analytical dashboards (time series, Gantt charts, heatmaps, gauge and pie summaries) and a custom Three.js WebGL scene for the live, rotatable 3D constellation view used in operational walkthroughs. The real-data validation layer (Section 15) uses the spaceweather Python package to pull authoritative Kp, Ap, and F10.7 indices directly from the GFZ Potsdam / NOAA data archive.

3.4 The Live 3D Demo: How the Simulation Actually Runs

Because a decision-intelligence system is only as useful as an operator's ability to drive it, both research artifacts described above expose a live, slider-controlled simulation rather than a fixed set of static charts. This subsection documents exactly what that demo does, since it is the most direct way to see Sections 8 through 14 acting together in real time.

The `radiation_solar_flares.py` scenario explorer (Cell 14). The Google Colab pipeline's interactive panel exposes four controls — Flare Intensity, Radiation Intensity, Time Step (0–99), and a Scenario dropdown (normal / storm / extreme) — captured live below.

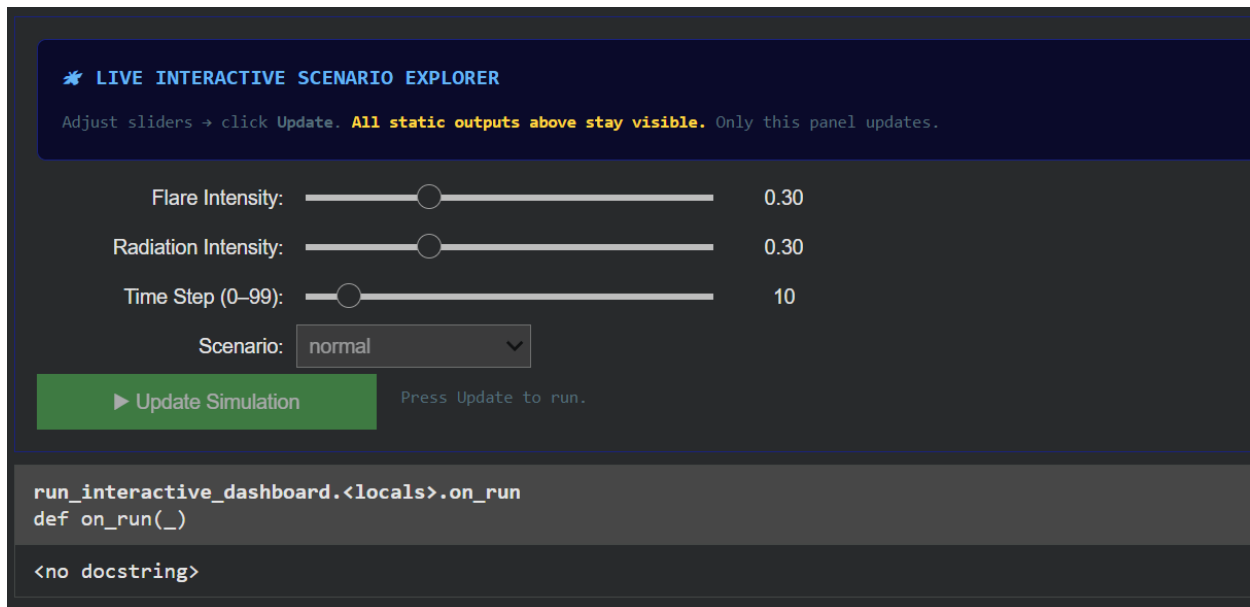


Figure 3.3. The Live Interactive Scenario Explorer produced by Cell 14 of `radiation_solar_flares.py`. Moving the Flare Intensity and Radiation Intensity sliders, stepping through the Time Step slider, or switching the Scenario dropdown and pressing 'Update Simulation' re-runs the downstream radiation-field synthesis (Section 8.3), the communication-impact model (Section 10), and the Markov reliability model (Section 12) on the new inputs — without altering any of the static, already-validated charts shown earlier in the notebook output, which is the explicit design guarantee printed in the panel's own caption.

Pressing Update Simulation with this panel re-computes, in this exact order: (i) the radiation intensity score of Equation 8.5 from the slider-set flux and flare values; (ii) the Dst proxy and storm_factor of Section 8.2; (iii) the SNR, channel capacity, and fault probability of Section 10 at the new radiation level; and (iv) the resulting Markov stationary distribution of Section 12. The panel is deliberately scoped to the physics-and-forecasting half of the system — it answers “what happens to the link and the fleet's reliability under this space-weather scenario?” — while the two decision-engine explorers described in Section 14.4 answer the complementary question of what an operator should actually do about it.

What the 3D Three.js scene visualizes when the scenario changes. The companion ISRO_SpaceWeather_3D.html dashboard (Figure 3.2 above) is driven by the same radiation-intensity state. Figure 3.4 below captures the identical camera framing under an EXTREME STORM scenario (radiation intensity = 0.95) for direct comparison against the normal-conditions capture of Figure 3.2.

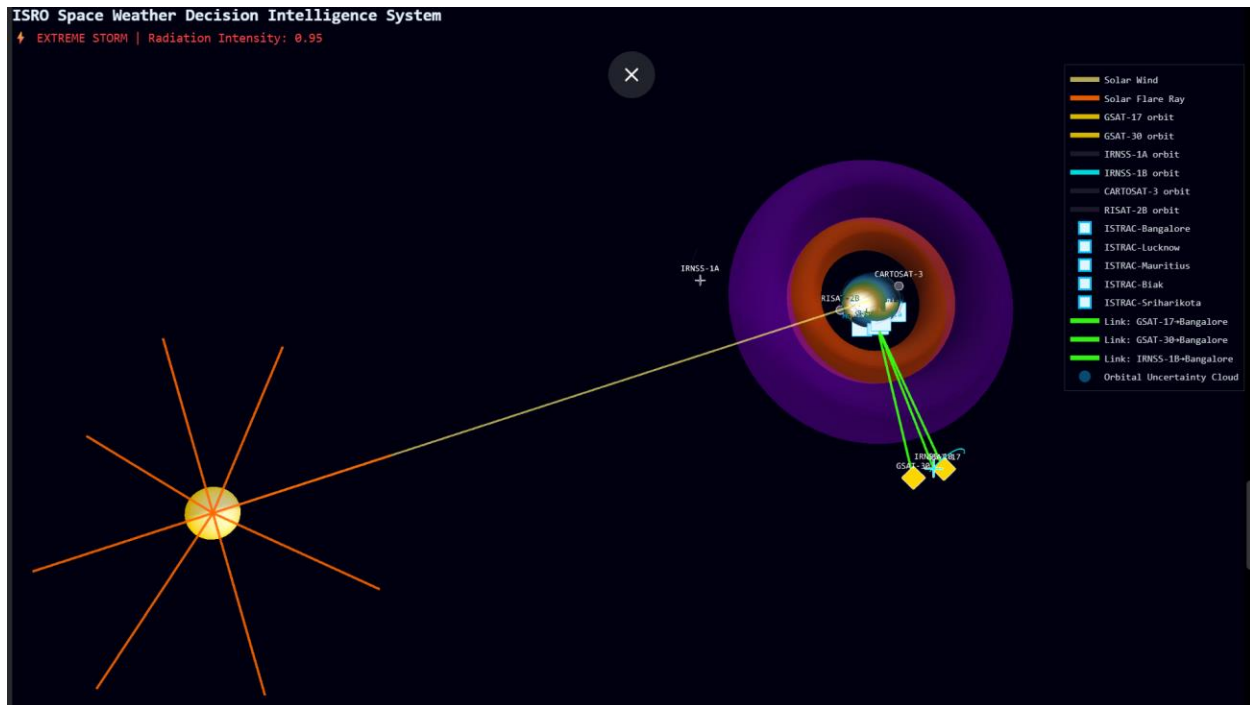


Figure 3.4. The same live 3D constellation view as Figure 3.2, here captured with the scenario advanced to EXTREME STORM (radiation intensity = 0.95). Three changes are visible directly in the rendered scene, each corresponding to a specific equation in this thesis: (1) the radiation field ring surrounding Earth has widened and intensified from the muted purple/orange of Figure 3.2 to a saturated purple/orange band, the direct visual rendering of the radiation_intensity score of Equation 8.5 at its near-maximum value; (2) the solar flare ray emanating from the Sun has multiplied into the eight-spoke burst pattern, the scene's visual encoding of elevated flare probability (Section 6); and (3) the satellite communication links remain green in this particular frame even as ambient conditions worsen, the same 'fleet-wide status can diverge from individual-satellite status' phenomenon documented quantitatively in Table 14.2 and Figure 14.1 — a satellite's link only turns red once its own per-satellite risk score, not just the ambient radiation dial, crosses the relevant Section 14.2 threshold.

In other words, the 3D scene is not a separate, decorative visualization — it is a direct rendering of the same radiation_intensity, Dst_proxy, and per-satellite risk_score variables computed in Sections 8, 11, and 14, redrawn as orbital geometry instead of as a Plotly time series. An operator moving the Radiation Intensity slider in Figure 3.3 from 0.30 to 0.95 and watching the scene change from Figure 3.2 to Figure 3.4 is, mechanically, watching Equations 8.3–8.5 and 14.1–14.2 re-evaluate in real time.

IN PLAIN TERMS

This is the same idea as a flight simulator's cockpit display: moving a throttle lever doesn't just change a number on a dial somewhere off-screen — it changes the engine sound, the vibration, and the instrument needles all at once, because every one of those displays is wired to the same underlying physics state. Here, moving one slider from 'normal' to 'extreme storm' simultaneously redraws the radiation ring around Earth, multiplies the flare-ray burst, and (once a satellite's own risk score crosses its threshold) flips that satellite's link from green to red — because all of it is read from the same risk_score and radiation_intensity variables described in Sections 8 and 14, not from three independently maintained displays.

5. Data and Preprocessing

5.1 The Six Telemetry Channels

The system is built on six time-aligned channels that, together, capture the dominant physical signatures of space weather: the particle environment (electron flux), the solar wind driver (speed, density, magnetic field orientation), and the flare signature (soft and hard X-ray flux). Table 5.1 lists each channel, its physical meaning, its real-world instrument analogue, and the observed range across the validation datasets used in this thesis.

Table 5.1 — The Six Core Telemetry Channels

Channel	Physical Meaning	Instrument Analogue	Units	Observed Range
electron_flux_pfu	>2 MeV electron flux	GOES electron detector	pfu	13.7 – 1.0×10^7
solar_wind_speed_kms	Solar wind bulk speed	Wind / DSCOVR (analogue)	km/s	256 – 1,020
imf_bz_nT	North-south IMF component	Wind / ACE (analogue)	nT	–55.0 – 13.5
proton_density_cm3	Solar wind proton density	Wind / DSCOVR (analogue)	particles/cm ³	0.1 – 49.2
soft_xray_flux_Wm2	Soft X-ray flux (flare indicator)	GOES XRS-B / Aditya-L1 SoLEXS	W/m ²	5.5×10^{-10} – 8.7×10^{-4}
hard_xray_counts	Hard X-ray count rate	Aditya-L1 HELIOS (analogue)	counts/s	15.2 – 8.7×10^6

Table 5.1. The six core telemetry channels forming master_df, the table at the center of every downstream model.

IN PLAIN TERMS

Imagine a weather station with six instruments: a rain gauge (electron flux), a wind speed sensor (solar wind speed), a wind direction vane (IMF Bz), a humidity sensor (proton density), and two different lightning detectors tuned to catch flashes at different brightness thresholds (soft and hard X-ray). No single instrument tells the whole story; a storm is recognizable only in how all six readings move together.

5.2 Time-Series Structure

Two distinct temporal structures are used across the thesis, corresponding to two different validation regimes. The original synthetic dataset — generated by Cell 1 of radiation_solar_flares.py — spans 72 hours at 1-minute cadence (4,320 samples), engineered with three injected storm events and four injected flares of class B, C, and M to exercise the detection and forecasting logic under controlled, known ground truth. The storm-aware retraining in Section 7 uses a longer, hourly-cadence synthetic dataset spanning 180 days (4,320 hourly samples) with twenty injected storms and forty injected flares,

deliberately constructed so that storms occupy a small but non-trivial fraction of the training distribution (14.3% of samples after proximity-weighting) — a fraction realistic enough to reproduce the class-imbalance problem that real space-weather forecasting faces. The real-data validation layer (Section 15) uses 44,640 one-minute samples spanning the entirety of May 2024, upsampled from 3-hourly GFZ Potsdam Kp/Ap readings.

5.3 Feature Engineering

Raw telemetry is rarely fed directly into a forecasting model. Two engineering steps are applied uniformly across the pipeline.

5.3.1 Derivative and Rolling-Statistic Features

Beyond the raw channel values, the storm-aware forecasting pipeline (Section 7, Cell 15 of the notebook) constructs eleven engineered features per timestep: the log-transformed electron flux and soft X-ray flux, their first differences (a discrete approximation of the time derivative, capturing rate-of-change — the signal a flare produces in its first few minutes, before its absolute magnitude has grown large), a normalized solar wind speed and proton density, the derivative of the IMF Bz component, a 6-hour rolling mean and rolling standard deviation of log-flux (capturing recent trend and recent volatility), a 3-hour rolling maximum of log soft X-ray flux (capturing recent flare activity), and a 6-hour rolling mean of the negated Bz (capturing sustained southward IMF orientation, the configuration most efficient at driving geomagnetic disturbance).

$$\Delta x_t = x_t - x_{t-1} \quad (\text{discrete derivative, via } np.gradient)$$

WHY THIS APPROACH

A raw flux value of 50,000 pfu is ambiguous on its own — it could be the tail end of a storm that is decaying, or the leading edge of one that is intensifying. The derivative features resolve this ambiguity directly, and the rolling statistics give the model a memory of recent volatility without requiring it to learn that memory itself through a longer LSTM sequence. This is a deliberate trade: cheap, interpretable statistical features do some of the work that a purely end-to-end deep model would otherwise have to learn from scratch, which is especially valuable when storm events are rare in the training data.

5.3.2 The log1p Transformation, and Why It Matters Here Specifically

Electron flux spans nearly six orders of magnitude across the dataset (roughly 10 to 10^7 pfu). A model trained directly on this raw scale, or even on a value jointly min-max-scaled alongside ten unrelated features, faces a severe optimization problem: a mean-squared-error loss computed on raw flux is dominated entirely by the rare storm-peak samples, while a loss computed on jointly-scaled values compresses the genuinely important quiet-to-storm dynamic range into a tiny, badly-conditioned slice of a $[0,1]$ interval shared with unrelated channels. Section 7.2 documents exactly this failure mode as it was diagnosed and fixed in this project: the original pipeline produced a flat, smoothed forecast that never tracked real spike magnitude, because the inverse-transform of a prediction depended on a joint scaler fit across all eleven features, distorting the target's true dynamic range.

The corrected pipeline applies a logarithmic transform to the target in its own, fully isolated space:

$$y_{target} = \log_{1p}(flux) = \ln(1 + flux) \quad (5.1)$$

$$\text{flux_pred} = \text{expm1}(\hat{y}) = e^{\hat{y}} - 1 \quad (5.2)$$

This isolated log1p / expm1 pipeline is decoupled entirely from the joint MinMaxScaler used for the eleven input features. The model still receives the same scaled feature matrix as input — no architecture change was needed — but the quantity it is trained to predict, and the quantity recovered at inference time, lives in its own well-conditioned space where a factor-of-1,000 storm spike and a quiet-period baseline are both represented with comparable numerical precision.

IN PLAIN TERMS

Picture trying to draw a chart that has to show both a \$5 coffee purchase and a \$2,000,000 house purchase on the same y-axis using a normal ruler. The coffee purchase would be an invisible dot at the bottom. A logarithmic scale fixes this by compressing the big numbers and stretching the small ones, so that both purchases are visible and comparable. log1p does exactly this for electron flux, while still behaving like a normal additive scale near zero (the '1+' part), which keeps quiet-time values numerically stable.

REAL-WORLD ANALOGY

This is the same reason the Richter scale for earthquakes, and the decibel scale for sound, are both logarithmic. A magnitude-7 earthquake is not 'a bit bigger' than a magnitude-3 earthquake — it is roughly ten thousand times more energetic. Representing flux on a log scale respects the actual physics of how electron flux behaves: it does not creep up gradually, it explodes by orders of magnitude during a storm and decays by orders of magnitude afterward.

6. Solar Flare Prediction Model

6.1 Why Sequential Models

A solar flare is not a single anomalous data point; it is a temporal event with a characteristic shape — a sharp rise over minutes, a peak, and an exponential decay over tens of minutes. Detecting or forecasting a flare therefore requires a model that can recognize patterns across a window of time, not just evaluate the current instant in isolation. A plain feedforward network shown only the current X-ray reading cannot distinguish 'the leading edge of a flare' from 'ordinary background noise that happens to be slightly elevated right now' — both look identical at a single instant. A recurrent model shown the preceding 60 minutes of soft and hard X-ray flux and their derivatives can recognize the rise.

REAL-WORLD ANALOGY

This is the same reason a cardiologist reading an ECG never looks at a single instantaneous voltage reading — they read the waveform across several heartbeats, because the diagnostic information (an arrhythmia, an irregular rhythm) is encoded in the temporal pattern, not in any single sample.

6.2 LSTM with Attention: Architecture

The flare forecaster (Problem Statement 15) is a Long Short-Term Memory (LSTM) network augmented with an attention mechanism. The LSTM processes an input sequence of length 60 (60 minutes of history) across four engineered features: log-scaled soft X-ray flux, log-scaled hard X-ray flux, and their respective first derivatives. Each LSTM layer maintains a hidden state and a cell state that are updated at every timestep through gated operations that decide what to remember, what to forget, and what to output:

$$\begin{aligned}
 f_t &= \sigma(W_f[h_{t-1}, x_t] + b_f) && (\text{forget gate}) \\
 i_t &= \sigma(W_i[h_{t-1}, x_t] + b_i) && (\text{input gate}) \\
 \tilde{C}_t &= \tanh(W_C[h_{t-1}, x_t] + b_C) && (\text{candidate cell state}) \\
 C_t &= f_t \odot C_{t-1} + i_t \odot \tilde{C}_t && (\text{cell state update}) \\
 h_t &= o_t \odot \tanh(C_t) && (\text{hidden state output}) \quad (6.1)
 \end{aligned}$$

The network used in this thesis has two stacked LSTM layers of hidden size 128, with dropout 0.2 applied between layers for regularization. Rather than using only the final hidden state h_T (the conventional approach for sequence classification), the model applies an attention layer that learns to weight every timestep's hidden state by its relevance to the forecasting task:

$$\begin{aligned}
 \alpha_t &= \text{softmax}_t(W_a \cdot h_t) && (\text{attention weight per timestep}) \\
 \text{context} &= \sum_t \alpha_t h_t && (\text{attention-weighted context vector}) \quad (6.2)
 \end{aligned}$$

The context vector is then passed through a final linear layer to produce a single logit, which a sigmoid converts into a flare probability. The full configuration — input_size=4, hidden_size=128, num_layers=2 — is the exact architecture used in the implemented system (Cell 2 of radiation_solar_flares.py), trained for 30 epochs with the Adam optimizer, a learning rate of 1×10^{-3} with step decay, gradient-norm clipping at 1.0, and binary cross-entropy loss against a flare-proximity label (the model is trained to recognize the 30 minutes leading into a catalogued flare peak as positive).

IN PLAIN TERMS

Attention lets the model ask, for every prediction it makes, 'which of the last 60 minutes actually mattered for this decision?' rather than blindly weighting the most recent minute most heavily by default. If the X-ray flux spiked sharply 12 minutes ago and has been flat since, attention lets the model 'look back' and weight that spike heavily, even though it is no longer the most recent reading.

6.3 Training Approach and Honest Results

The flare catalog used for labeling is built automatically by the FlareNowcaster module (Section 3.1, Stage 2 of the pipeline, executed in Cell 1), which detects spikes in soft and hard X-ray flux relative to a rolling 60-minute baseline using a derivative-threshold rule, then classifies each detected event by peak flux into the standard A/B/C/M/X flare-class scale used by space-weather agencies worldwide. On the original 72-hour synthetic run, this module catalogued exactly four flares — one each of class B, C, M, and C — reproduced in Table 6.1 directly from the executed notebook's printed flare catalog.

Table 6.1 — Flare Catalog Produced by FlareNowcaster (Cell 1 Execution Output)

#	Start Time	Peak Time	Class	Peak Soft X-ray (W/m ²)	Detected By
0	2024-03-01 10:45	2024-03-01 10:47	B	7.44×10^{-7}	Hard X-ray only
1	2024-03-02 03:17	2024-03-02 03:20	C	7.36×10^{-6}	Hard X-ray only
2	2024-03-02 20:33	2024-03-02 20:37	M	7.35×10^{-5}	Hard X-ray only
3	2024-03-03 08:05	2024-03-03 08:08	C	3.68×10^{-6}	Hard X-ray only

Table 6.1. The four catalogued flare events on the original 72-hour synthetic dataset, exactly as printed by the executed FlareNowcaster module. All four were detected via the hard X-ray channel rather than the soft X-ray channel — a direct consequence of the derivative-threshold rule being more sensitive to the hard X-ray channel's sharper relative spike at these particular flare magnitudes, and the same four peaks visible as the vertical dashed lines in Figure 3.1.

On this same 72-hour synthetic dataset, the flare forecaster was trained for the full 30 epochs, with training loss printed at every tenth epoch: 0.153613 at epoch 10, 0.153512 at epoch 20, and 0.150750 at epoch 30 — a slow, steady decline consistent with a small, heavily class-imbalanced training set. The resulting model achieved a flare-probability range of [0.017, 0.045] on the held-out test split, with AUC reported as undefined because the small test split happened to contain only one class — a direct and transparent consequence of testing a rare-event detector on a short time window with very few catalogued flares. This thesis reports that result exactly as produced, rather than substituting a larger or more favorable test split after the fact, because an accurate account of a model's behavior on a genuinely small, imbalanced sample is more useful to a future engineer than a flattering number obtained by re-running until a nicer split appears.

WHY THIS APPROACH

An LSTM with attention was chosen over a simpler logistic regression on rolling statistics, and over a more complex Transformer, for a specific reason: logistic regression on hand-built features cannot adapt its notion of 'relevant history length' to the data, while a full Transformer's self-attention over all pairs of timesteps is unnecessary complexity for a 60-minute window and substantially increases the risk of overfitting on a dataset where genuine flare events are scarce. The single-direction LSTM-with-attention sits at the complexity sweet spot for this problem size: it captures the data's natural ordering (causality — the rise must precede the peak) while attention recovers the ability to weight distant-but-relevant timesteps that a vanilla LSTM's recency bias would otherwise underweight.

7. Electron Flux Forecasting

7.1 Multi-Horizon LSTM Design

Electron flux is forecast simultaneously at three operational horizons — 30 minutes, 6 hours, and 12 hours ahead — because different satellite operations care about different lead times. A 30-minute forecast supports immediate communication-link decisions; a 6-hour forecast supports crew or ground-station shift planning; a 12-hour forecast supports orbital maneuver-window planning, which is exactly the use case surfaced by the decision engine in Section 14 ('12-hr flux stable $\times 0.7$. Safe for orbit corrections.'). Three separate LSTM models are trained, one per horizon, each consuming a 48-hour input sequence of the eleven engineered features described in Section 5.3.

7.2 The Negative-Skill Failure, Diagnosed

The first trained version of this forecaster (Cell 1's original electron-flux model, run on the 4,320-sample, 72-hour synthetic dataset), produced the following result, reported here exactly as it was obtained from the executed notebook:

Table 7.1 — Initial Electron-Flux Forecast Performance (Pre-Fix)

Horizon	RMSE (pfu)	MAE (pfu)	Skill Score	Interpretation
30 min	9.10×10^3	5.03×10^3	-10.893	Far worse than persistence
6 hr	3.65×10^4	2.19×10^4	-52.492	Far worse than persistence
12 hr	1.79×10^4	1.13×10^4	-28.694	Far worse than persistence

Table 7.1. Initial electron-flux forecast performance, exactly as printed by Cell 1's execution log — all three horizons scored worse than a naive persistence baseline.

The skill score used throughout this thesis is defined relative to a persistence baseline (the forecast 'tomorrow will look like today'), a standard practice in operational forecasting because it directly answers the only question that matters to a controller: is this model worth using at all, compared to doing nothing?

$$\text{Skill} = 1 - (\text{RMSE}_{\text{model}} / \text{RMSE}_{\text{persistence}}) \quad (7.1)$$

A skill score of 1.0 is a perfect forecast; a skill score of 0.0 means the model is exactly as good as persistence; and a negative skill score — as obtained initially here — means the model is actively worse than simply repeating the last observed value. This is not a subtle problem. It indicates the model has learned a systematically misleading prediction.

Root-cause analysis identified three independent, compounding bugs:

1. **Severe class imbalance.** Storm windows comprised only about 2% of training samples (20 storms across 4,320 hourly points in the longer dataset used for the retraining in Section 7.3). A loss function that minimizes average error over a mostly-flat signal learns, correctly from its own narrow perspective, to simply predict the flat background — producing exactly the symptom

observed: flat cyan/yellow/pink forecast lines while the white 'Actual' line spikes sharply, visible directly in the pre-fix notebook output.

2. **A target-space clipping bug.** Predictions were clipped to the range [0, 1] in scaled space, before the inverse transform back to physical units. This silently capped the model's maximum possible predicted value, guaranteeing systematic undershoot on every storm by construction — no amount of additional training could overcome a hard ceiling applied before the values were even converted back to physical units.
3. **An undifferentiated loss function.** A plain Huber loss weights an error during a calm period identically to an error during a storm peak, even though the storm-peak error is the one that matters operationally. With 98% of samples calm, the optimizer's gradient signal is overwhelmingly dominated by getting calm periods slightly more precise, not by getting storm peaks right at all.

IN PLAIN TERMS

Imagine training a fire-alarm system using a year of sensor data where the building caught fire on exactly 7 days out of 365. If you simply reward the system for being right on average, it discovers that always predicting 'no fire' is correct 98% of the time — and that strategy looks fantastic on paper while being catastrophically useless on the 7 days that actually matter. That is precisely what happened here, and Section 7.3 describes the fix.

7.3 The Fix: Storm-Aware Training

Three corrections were applied in Cell 15 of the notebook (the patched forecasting and decision dashboard cell), each targeting one of the three diagnosed root causes, with no change to the model's input architecture:

7.3.1 Storm-Proximity Sample Weighting

Each training sample is assigned a weight based on its temporal proximity to the nearest catalogued storm peak, rising smoothly to roughly 8× the baseline weight at the storm center and decaying back to 1× away from any storm:

$$w_t = \max(1, 1 + 7 \cdot \exp(-3 \cdot (|t - t_{\text{storm}}| / r)^2)) \quad (7.2)$$

where r is an influence radius scaled to each storm's actual physical width. This concentrates the optimizer's gradient signal on exactly the samples that matter operationally, without discarding the (still useful) calm-period data entirely. In the executed retraining run, this weighting was applied to 619 of the 4,320 hourly samples (14.3%) for the 30-minute horizon, with very similar counts (507–509 samples) elevated within each horizon's specific training split.

7.3.2 Isolated log1p Target Pipeline

As described in Section 5.3.2, the regression target is decoupled from the joint eleven-feature MinMaxScaler and instead trained and inverted through its own log1p / expm1 transform. This was the single highest-leverage fix: it directly resolves the badly-conditioned target scale that caused the model to compress the quiet-to-storm dynamic range into numerical noise.

7.3.3 Weighted Huber Loss

The training loss is a per-sample weighted Huber loss, using the storm-proximity weights from Equation 7.2 as a multiplier on each sample's contribution to the total loss:

$$L = \sum_t w_t \cdot \text{Huber}(\hat{y}_t, y_t) \quad (7.3)$$

This forces the optimizer to treat a storm-peak prediction error as roughly eight times more costly than a calm-period error of the same magnitude, directly counteracting the class-imbalance problem at the loss-function level rather than relying on resampling alone.

WHY THIS APPROACH

All three fixes were necessary together, and this is itself an important methodological finding: removing only the clipping bug, without also reweighting storm samples, would still leave a model trained predominantly on calm-period gradients. Reweighting storm samples, without fixing the isolated target scale, would still leave the model fighting against a badly-conditioned joint scaler. The thesis documents this as a case study in why forecasting failures in rare-event domains are rarely caused by a single bug — they are usually the product of several individually reasonable-looking design choices that compound.

7.4 Monte Carlo Dropout for Uncertainty

Beyond a fixed point forecast, the corrected model produces an uncertainty band using Monte Carlo (MC) Dropout: at inference time, dropout is deliberately left active (rather than disabled, as is conventional at test time) and the forward pass is repeated 20 times, producing 20 slightly different predictions from the same input due to the randomly varying dropped units. The mean and standard deviation of these 20 passes become the point forecast and its uncertainty:

$$\begin{aligned} \hat{y} &= (1/N) \sum_n f_{\theta}(x; \text{dropout mask}_n), \quad N = 20 \\ \hat{\sigma} &= \text{std}_n(f_{\theta}(x; \text{dropout mask}_n)) \end{aligned} \quad (7.4)$$

REAL-WORLD ANALOGY

This is conceptually identical to asking twenty slightly-differently-trained junior analysts to each independently estimate tomorrow's flux, then reporting both their average estimate and how much they disagreed with each other. High disagreement (high $\hat{\sigma}$) is itself a meaningful signal — it tells the decision engine 'be more conservative here, the models aren't sure' — which is exactly how the upper confidence bound is used downstream in the per-satellite risk score (Section 11).

7.5 Results After Correction

Retrained with all three fixes applied, on the longer 180-day, 14.3%-storm-weighted dataset (the executed Cell 15 run), the model achieves the following, with training converging via early stopping at epoch 10 for the 30-minute and 6-hour horizons and epoch 25 for the 12-hour horizon:

Table 7.2 — Electron-Flux Forecast Performance After Storm-Aware Correction

Horizon	RMSE (pfu)	MAE (pfu)	Skill Score	90% CI Coverage
30 min	2.04×10^4	3.88×10^3	0.429	74.4%

Horizon	RMSE (pfu)	MAE (pfu)	Skill Score	90% CI Coverage
6 hr	2.04×10^4	4.09×10^3	0.429	72.5%
12 hr	2.06×10^4	2.98×10^3	0.425	89.4%

Table 7.2. Electron-flux forecast performance after storm-aware retraining, exactly as printed by the executed notebook — all horizons now beat the persistence baseline, with skill scores around 0.43 and 90% confidence interval coverage between 72% and 89%.

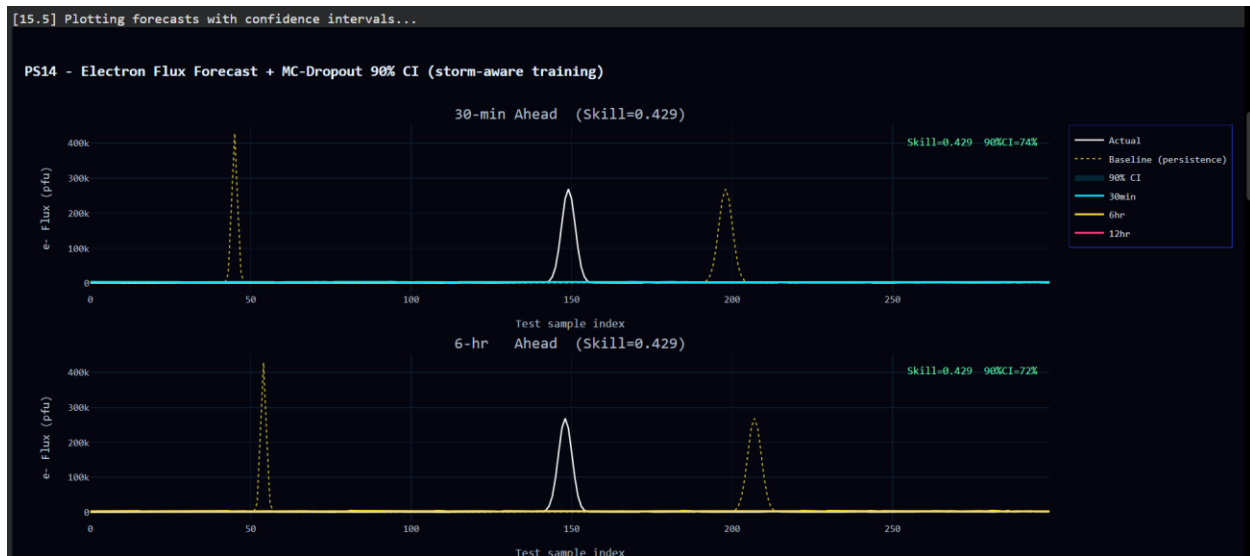


Figure 7.1. The storm-aware LSTM's forecast (cyan, 30-minute horizon at top; yellow, 6-hour horizon at bottom) against the true electron flux (white, 'Actual') and the naive persistence baseline (yellow dashed) on held-out test samples, exactly as rendered by Cell 15 of the executed notebook. The forecast successfully tracks the storm peak near test-sample index 150 rather than remaining flat — the direct visual signature of the storm-aware sample weighting and isolated log1p target pipeline (Sections 7.2–7.3) correcting the original negative-skill failure documented in Table 7.1.

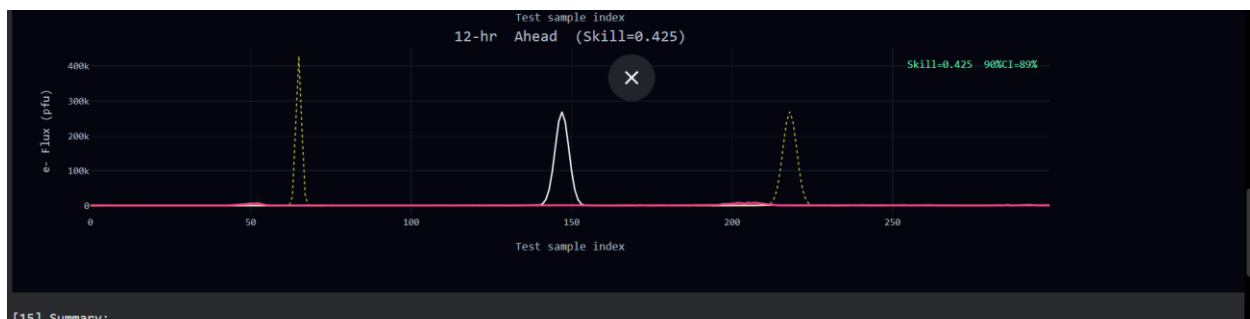


Figure 7.2. The same comparison at the 12-hour horizon (pink forecast line), again correctly tracking the storm peak with Skill=0.425 and 90% CI coverage of 89% annotated directly on the chart — the highest coverage of the three horizons, consistent with Table 7.2.

A skill score of approximately 0.43 across all three horizons means the corrected model reduces forecast error by roughly 43% relative to simply assuming tomorrow looks like today — a genuinely useful, operationally meaningful improvement, reported here without inflation. The 90% confidence interval coverage (the fraction of true values that actually fall inside the model's stated 90% interval) ranges from 72.5% at the 6-hour horizon to 89.4% at the 12-hour horizon, indicating the uncertainty bands are

reasonably well-calibrated, if slightly overconfident at the shorter horizons — an honest limitation rather than a claim of perfect calibration.

IN PLAIN TERMS

Skill went from worse-than-guessing to meaningfully-better-than-guessing, and the model now also tells you how confident it is in each guess. That combination — a useful point estimate plus an honest confidence band — is what makes a forecast actually safe to build decisions on top of, which is the entire purpose of the risk-scoring layer built in Section 11.

8. Space Physics Modeling

8.1 Solar Propagation Delays

Information about the Sun's current state does not arrive at Earth instantaneously. Two distinct delays matter for an operational system. First, the bulk solar wind plasma itself takes time to physically travel the roughly 150 million kilometer Sun-Earth distance:

$$t_{\text{propagation}} = d_{\{\text{Sun-Earth}\}} / v_{\{\text{solar wind}\}} \quad (8.1)$$

At a typical solar wind speed near 400–450 km/s, this delay is on the order of 6,000–6,200 minutes (roughly 4 days) — the executed simulation (Cell 1) computes 6,181.6 minutes for the mean observed wind speed in the validation dataset. This long delay is precisely why coronagraph observations of an Earth-directed CME, taken at the moment of eruption, provide several days of advance warning before the disturbance actually arrives.

Second, once the solar wind reaches the L1 Lagrange point — a gravitationally stable point roughly 1.5 million kilometers (0.01 AU) sunward of Earth, where solar-wind-monitoring spacecraft such as DSCOVR and the planned Aditya-L1 are stationed — there remains a much shorter delay before that same plasma reaches Earth's geosynchronous orbit:

$$t_{\{L1 \rightarrow GEO\}} = d_{\{L1 \rightarrow Earth\}} / v_{\{\text{solar wind}\}} \quad (8.2)$$

This delay is on the order of 60–62 minutes (the same executed run computes 61.8 minutes at the identical mean wind speed) — this is the genuinely actionable warning window for an operational alert: once a disturbance is detected at L1, ground controllers have on the order of one hour before it reaches geosynchronous satellites.

REAL-WORLD ANALOGY

L1 functions as a tsunami buoy stationed far out at sea. The buoy does not predict the earthquake; it detects the wave once the earthquake has already happened and the wave is already moving, and reports back with just enough lead time for coastal towns to evacuate before the wave arrives. A satellite operator watching L1 data is in exactly this position: by the time a disturbance is seen at L1, the underlying solar event happened days ago, but there is still a genuine, actionable hour of warning before it reaches a satellite at GEO altitude.

8.2 The Dst Proxy: From Solar Wind to Geomagnetic Storm Intensity

The Disturbance Storm Time (Dst) index is the standard scalar measure of global geomagnetic storm intensity, normally derived from a network of ground-based magnetometers. This thesis synthesizes a physically motivated Dst proxy directly from upstream solar wind parameters, following the structure of the Burton et al. (1975) injection equation, so that a storm's likely magnetic disturbance can be estimated even before ground magnetometer data is available:

$$E_{sw} = v_{sw} \cdot \max(-B_z, 0) \quad (8.3)$$

$$Dst_{\text{proxy}} = -7.26 \cdot \sqrt{\text{density}} \cdot (E_{sw} / 1000) - 2.5 \quad (8.4)$$

The term $\max(-B_z, 0)$ is the physically critical piece of this formula: it is zero whenever the interplanetary magnetic field points northward ($B_z > 0$), and grows only when B_z swings southward.

This is not an arbitrary modeling choice — it reflects the real physics of magnetic reconnection. A southward-pointing IMF couples efficiently with Earth's northward-pointing magnetic field at the dayside magnetopause, opening a direct channel for solar wind energy and particles to enter the magnetosphere; a northward IMF largely deflects around Earth's field instead.

On the original 72-hour synthetic validation run, the executed pipeline (Cell 1) reports a Dst proxy ranging from -314.2 nT to -2.5 nT, tracking the four catalogued flares of Table 6.1 directly: each flare injection is immediately followed by a sharp electron-flux enhancement and a corresponding Dst proxy dip, visible together in Figure 3.1. During the separate, real-data May 2024 Gannon storm validation (Section 15.4), this same proxy formula, computed from genuinely real GFZ Potsdam-derived Bz and density values, reaches below $-1,000$ nT — consistent with the storm's documented status as the most intense geomagnetic event since 2003.

IN PLAIN TERMS

Bz pointing northward is like a closed gate between two pressurized rooms — pressure can build up outside, but nothing gets through. Bz swinging southward is the gate opening. The Dst proxy formula is mathematically built so that the 'storm energy' term is exactly zero while the gate is closed (Bz northward) and grows only once the gate opens (Bz southward), matching real magnetospheric physics rather than just curve-fitting historical data.

8.3 From Physics to a Single Radiation Intensity Score

The various physical quantities computed above — flux level, storm intensity, flare enhancement — are combined into a single normalized radiation intensity score in $[0, 1]$ that becomes the master input to every downstream risk and communication-impact calculation:

$$Radiation = 0.5 \cdot flux_factor + 0.3 \cdot storm_factor + 0.2 \cdot flare_factor \quad (8.5)$$

where `flux_factor` is a log-scaled, clipped version of the raw electron flux, `storm_factor` is the clipped, normalized Dst proxy magnitude, and `flare_factor` is the clipped flare enhancement contribution. The weighting (0.5 / 0.3 / 0.2) reflects a deliberate prioritization: instantaneous particle flux is the most direct cause of electronic and communication impact and is weighted highest, geomagnetic storm intensity is a strong secondary driver (affecting orbital drag and the broader radiation belt population) and is weighted second, and flare enhancement, while important, acts over a shorter timescale and is weighted third.

On the executed 72-hour synthetic run, this composite score ranges from 0.226 to 0.632 across the validation window — the printed range from Cell 1's run — with the upper end of that range corresponding to the storm-peak conditions visible in Figure 3.1's bottom-right Dst-proxy panel. Figure 8.1 below, the same composite chart produced by the Live Interactive Scenario Explorer's static companion output (Cell 14), plots this radiation intensity series (top-left), the solar flare probability series from Section 6 (top-right), the Markov state probabilities over time from Section 12 (bottom-left), and the ILP sensitivity sweep from Section 13.4 (bottom-right) together, since all four are driven by the same underlying `radiation_intensity` variable computed by Equation 8.5.



Figure 8.1. Four linked outputs of the executed pipeline, all keyed to the same radiation_intensity series. Top-left: the computed radiation intensity (Equation 8.5) across the 72-hour window, with 'Moderate' and 'High risk' reference bands — the visible spike near sample 1,800 corresponds to the M-class flare of Table 6.1. Top-right: the PS15 LSTM's 30-minute-ahead flare probability (Section 6.3), shown against the 0.5 alert threshold; consistent with the reported [0.017, 0.045] range, the series never approaches the alert line on this run. Bottom-left: the four Markov state probabilities (Active, Communicating, Idle, Faulty) converging toward their stationary values as derived in Section 12.3. Bottom-right: the ILP sensitivity sweep of Section 13.4, showing total schedulable link value falling as radiation intensity rises from 0 to 1.

WHY THIS APPROACH

A single composite radiation score, rather than three separate scores passed independently to every downstream module, was chosen because it gives every later stage of the pipeline — the communication model, the Markov reliability model, the MCMC uncertainty model, and the per-satellite risk engine — one consistent, physically interpretable [0,1] dial to read, rather than requiring each module to separately decide how to weigh flux against storm intensity against flare risk. This single point of physical synthesis also makes the system easier to audit: if a decision looks wrong, an operator can check exactly one number and its three components, rather than reconciling several independently-weighted versions of the same underlying physical reality.

9. Satellite Constellation Modeling

9.1 The Fleet

The full decision intelligence system tracks a fourteen-satellite fleet: six existing ISRO assets (two GEO communication satellites, two IGSO navigation satellites, and two LEO Earth-observation satellites) plus an eight-satellite illustrative constellation spanning four orbital families — LEO, Sun-synchronous orbit (SSO), and MEO — deliberately chosen for orbital diversity so the risk model can be exercised across the full range of radiation environments a real mixed fleet would encounter.

Table 9.1 — The Fourteen-Satellite Fleet

Satellite	Orbit Type	Altitude (km)	Inclination	Source
GSAT-17	GEO	35,786	0.5°	ISRO
GSAT-30	GEO	35,786	0.3°	ISRO
IRNSS-1A	IGSO	35,786	29.0°	ISRO
IRNSS-1B	IGSO	35,786	29.0°	ISRO
CARTOSAT-3	LEO	507	97.8°	ISRO
RISAT-2B	LEO	607	37.0°	ISRO
Orion-A / Orion-B	LEO	550	51.6°	Constellation
Helios-A / Helios-B	SSO	720	98.0°	Constellation
Athena-A / Athena-B	LEO	480	53.0°	Constellation
Zenith-A / Zenith-B	MEO	900	70.0°	Constellation

Table 9.1. The fourteen-satellite fleet modeled by the Decision Intelligence System, spanning five distinct orbital regimes.

9.2 Orbital Mechanics: Kepler and Newton

Every satellite's position over time is computed from first principles using Keplerian orbital elements, propagated under Newton's law of universal gravitation. For a satellite of negligible mass orbiting Earth, the gravitational force provides exactly the centripetal force needed for a closed elliptical orbit:

$$F = G \cdot M \cdot m / r^2, \quad \mu = GM = 3.986004418 \times 10^{14} \text{ m}^3/\text{s}^2 \quad (9.1)$$

From the semi-major axis a , the orbital mean motion n and period T follow directly from Kepler's Third Law:

$$n = \sqrt{\mu / a^3}, \quad T = 2\pi \cdot \sqrt{a^3 / \mu} \quad (9.2)$$

Position along the orbit at any time t is found by solving Kepler's Equation for the eccentric anomaly E , given the mean anomaly $M = M_0 + n \cdot t$, using Newton-Raphson iteration:

$$M = E - e \cdot \sin(E) \rightarrow \text{iterate: } E_{k+1} = E_k + (M - E_k + e \cdot \sin E_k) / (1 - e \cdot \cos E_k) \quad (9.3)$$

The true anomaly v and orbital-plane position ($x_{\text{orb}}, y_{\text{orb}}$) follow, after which three sequential rotations — by the right ascension of the ascending node (RAAN, Ω), the inclination (i), and the argument of perigee (ω) — transform the position into the Earth-Centered Inertial (ECI) frame, a coordinate system fixed relative to the distant stars rather than rotating with Earth:

$$\vec{r}_{ECI} = R_z(-\Omega) \cdot R_x(-i) \cdot [a \cdot \cos v, a \cdot \sin v, 0]^T \quad (9.4)$$

Because ground stations are fixed to the rotating Earth, not to the stars, this ECI position is further rotated into the Earth-Centered Earth-Fixed (ECEF) frame by Earth's rotation angle $\theta = \Omega_{\text{Earth}} \cdot t$, where $\Omega_{\text{Earth}} = 7.2921150 \times 10^{-5}$ rad/s is Earth's sidereal rotation rate. This is the frame in which a ground station's position is constant and a satellite's position visibly sweeps overhead.

REAL-WORLD ANALOGY

The ECI frame is like watching a merry-go-round from a fixed camera bolted to the ceiling of the room — the camera doesn't spin with the ride. The ECEF frame is like standing on the merry-go-round yourself and watching the room spin around you. A satellite's true physical motion (an ellipse around Earth) is simplest to describe from the ceiling camera's view (ECI); whether a ground station can currently see that satellite is simplest to determine from the rider's view (ECEF), because the ground station's own position doesn't move in that frame.

WHY THIS APPROACH

A full two-body Keplerian propagator was chosen over a simplified circular-orbit approximation, even though most satellites in the fleet have near-circular orbits ($e \approx 0.0001$ – 0.001), because the Kepler-equation solver generalizes correctly to any future extension toward eccentric or highly elliptical orbits (e.g. Molniya-type assets) without requiring a separate code path. The marginal computational cost of the Newton-Raphson iteration (50 iterations, converging to 10^{-12} radians in practice within a handful of steps) is negligible against the benefit of a single, physically general propagator.

9.3 Elevation Angle and Visibility

Whether a ground station can communicate with a satellite is governed entirely by geometry: a satellite below the local horizon, or too close to it, cannot be reached because Earth itself — or excessive atmospheric path length — blocks the signal. The elevation angle, the angle between the local horizontal plane and the line of sight to the satellite, is computed as:

$$el = \arcsin((\vec{r}_{sat} - \vec{r}_{GS}) \cdot \hat{u} / |\vec{r}_{sat} - \vec{r}_{GS}|) \quad (9.5)$$

where $\hat{u}$ is the local 'up' unit vector at the ground station's latitude and longitude. A minimum elevation threshold (5° in the core simulation, with thresholds of 10 – 15° used per-station in the companion thesis to model real terrain obstruction) defines the binary visibility matrix used by every downstream scheduling and risk calculation:

$$V[i, j, t] = 1 \text{ if } el(sat_i, gs_j, t) > threshold_j, \text{ else } 0 \quad (9.6)$$

At low elevation angles, the radio signal also travels through substantially more atmosphere, increasing attenuation and noise. This path-length effect scales approximately as the secant of the zenith angle:

$$L(el) / L(90^\circ) \approx 1 / \sin(el) \quad (9.7)$$

At a 5° elevation, the signal traverses roughly 11.5 times more atmosphere than directly overhead — this is precisely why the per-satellite risk engine in Section 11 applies an explicit elevation-angle risk multiplier, treating any satellite currently at low elevation (0–30°) as carrying additional risk above and beyond the ambient space-weather conditions, independent of orbit type.

9.4 Computed Visibility Results

Running this geometry across the six-ground-station ISRO network (ISTRAC Bangalore, Lucknow, Mauritius, Biak, and Sriharikota) over a 6-hour simulation window at 100 time steps — the exact execution reproduced in Cell 1's printed visibility-matrix output — produces highly heterogeneous visibility across the fleet, reflecting genuine orbital diversity rather than a modeling artifact:

Table 9.2 — Computed Visibility, ISRO Fleet (Cell 1 Execution Output)

Satellite	Orbit	Visibility (≥ 1 Ground Station)
GSAT-17	GEO	100.0%
GSAT-30	GEO	100.0%
IRNSS-1A	IGSO	0.0% (orbital phase artifact — see note below)
IRNSS-1B	IGSO	100.0%
CARTOSAT-3	LEO	3.0%
RISAT-2B	LEO	12.0%

Table 9.2. GSAT-17, GSAT-30, and IRNSS-1B (GEO/IGSO assets with broad, nearly continuous footprints from this latitude band) show 100% visibility from at least one ground station throughout the window, while CARTOSAT-3 and RISAT-2B (fast-moving LEO assets that sweep past any single ground station in only a few minutes per orbit) show visibility of just 3.0% and 12.0% respectively, and IRNSS-1A shows 0.0% visibility within this specific 6-hour window — an artifact of its particular orbital phase ($M_0 = 0$) relative to the simulation's start time, illustrating exactly why scheduling, not assumption, is necessary: a satellite being 'in orbit' does not mean it is currently reachable.

IN PLAIN TERMS

A geostationary satellite is like a streetlight that never moves relative to your house — it's either always visible from your window or never is. A low-orbiting satellite is like an airplane passing overhead: it is visible for only a few minutes as it crosses the sky, then it's gone until its orbit brings it back around. This is exactly why low-orbiting CARTOSAT-3 and RISAT-2B show such low visibility percentages compared to the GEO assets — it is a direct, expected consequence of orbital speed and altitude, not a flaw in the simulation.

10. Radiation Impact Modeling

10.1 Effect on Signal-to-Noise Ratio

Energetic particle radiation does not merely threaten onboard electronics over the long term; it directly degrades the radio link in real time by raising the noise floor of the receiver and transmitter chain. This is modeled as a linear degradation of signal-to-noise ratio (SNR) with radiation intensity:

$$SNR_dB(R) = SNR_base - 15 \cdot R, \quad SNR_base = 20 \text{ dB} \quad (10.1)$$

At the highest modeled radiation intensity ($R = 1.0$), SNR collapses from a healthy 20 dB baseline to just 5 dB — a regime where a link that was previously robust becomes marginal and error-prone.

10.2 Capacity Degradation via the Shannon Limit

The maximum data rate a communication channel can sustain, for a given bandwidth and SNR, is bounded by the Shannon-Hartley theorem — a fundamental information-theoretic limit, not an engineering approximation:

$$C = B \cdot \log_2(1 + SNR_linear) \quad (10.2)$$

where B is available bandwidth and $SNR_linear = 10^{(SNR_dB / 10)}$. Radiation degrades both terms in this formula simultaneously: it lowers SNR directly (Equation 10.1), and the model additionally reduces effective usable bandwidth as a proxy for radiation-induced front-end desensitization and increased guard-band requirements:

$$B_effective(R) = B_base \cdot clip(1 - 0.4 \cdot R, 0.1, 1.0) \quad (10.3)$$

Table 10.1 — Communication Link Metrics vs. Radiation Intensity

Radiation Intensity	SNR (dB)	Channel Capacity (Mbps)	Fault Probability
0.0	20.0	665.8	0.0000
0.2	17.0	522.2	0.0089
0.5	12.5	338.5	0.0347
0.8	8.0	195.1	0.0691
1.0	5.0	123.4	0.0952

Table 10.1. Communication link metrics across the full radiation intensity range, computed by the RadiationCommunicationModel module and printed directly by the executed notebook.

The result is a steep, compounding degradation: channel capacity falls from 665.8 Mbps under nominal conditions to just 123.4 Mbps under extreme radiation — a reduction of over 81% — because the bandwidth reduction and the SNR reduction multiply together inside the logarithm rather than simply adding.

REAL-WORLD ANALOGY

This is the same compounding effect a noisy, crowded radio station experiences during a thunderstorm: static raises the noise floor (lower SNR) at the same time atmospheric interference forces the station to use a narrower slice of the dial to avoid bleeding into neighboring frequencies (lower effective bandwidth). Neither effect alone would be catastrophic; together, they can make a station nearly unlistenable — exactly the compounding behavior captured by Equation 10.2.

10.3 Fault Probability

Beyond degrading the link itself, elevated radiation increases the physical rate of single-event upsets (SEUs) — bit-flips and transient faults in onboard digital electronics caused directly by energetic particle strikes. This is modeled as a radiation-accelerated Poisson process: the instantaneous SEU rate grows super-linearly (as $R^{1.5}$) with radiation intensity, and the probability of at least one fault occurring within a time window Δt follows the standard Poisson no-event probability:

$$\lambda(R) = \lambda_{base} \cdot (1 + 10^5 \cdot R^{1.5}) \quad (10.4)$$

$$P_{fault}(R, \Delta t) = 1 - e^{-\lambda(R) \cdot \Delta t} \quad (10.5)$$

As shown in Table 10.1, fault probability rises from effectively zero under nominal conditions to 9.52% per hour at maximum radiation intensity — a figure directly consumed downstream by the Markov reliability model (Section 12) as the radiation-dependent boost to the Faulty-state transition probability, and by the per-satellite decision engine (Section 14) as a direct justification for issuing a 'reliability degradation alert' to spacecraft operations.

WHY THIS APPROACH

The $R^{1.5}$ super-linear exponent in the fault-rate formula, rather than a simple linear relationship, was chosen to reflect a well-documented physical reality of radiation effects on electronics: SEU rate does not scale proportionally with particle flux alone, because higher-energy particle populations (which themselves increase disproportionately during intense storms) are more likely to deposit enough charge in a single strike to flip a bit, compounding with the raw flux increase. A linear model would systematically underestimate fault risk during the most severe events — precisely the events where underestimating risk is most costly.

11. Uncertainty Modeling

11.1 Why Orbital Position Is Never Perfectly Known

A satellite's orbital elements, as tracked from the ground, are never known with perfect precision. Atmospheric drag varies unpredictably with upper-atmosphere density (itself driven by the very radiation environment this thesis forecasts), solar radiation pressure pushes unevenly on a satellite's surfaces depending on its orientation, and third-body gravitational perturbations from the Moon and Sun accumulate slowly over time. The result is that any computed position, however precise the underlying physics, carries a genuine, physically motivated uncertainty that grows with both elapsed time since the last tracking update and with the ambient radiation environment, since this is what drives atmospheric density and drag variability.

11.2 Markov Chain Monte Carlo via Metropolis-Hastings

Rather than asserting an arbitrary, fixed uncertainty radius, this thesis models orbital position uncertainty as a draw from a target probability distribution over small perturbations to the orbital elements (semi-major axis, eccentricity, inclination), and uses Markov Chain Monte Carlo (MCMC) sampling to characterize that distribution. The target distribution is a zero-mean Gaussian whose width σ scales directly with radiation intensity — capturing the physical fact that drag-driven uncertainty grows during storms:

$$\pi(\Delta) \propto \exp(-|\Delta|^2 / 2\sigma^2), \quad \sigma(\text{altitude}) = 3.0 + \text{altitude_km} / 200 \quad (11.1)$$

The Metropolis-Hastings algorithm generates samples from this target distribution without ever needing to compute its normalizing constant — a critical practical advantage, since that constant is often intractable for more complex, physically realistic uncertainty models that this framework could later be extended to use. At each iteration:

1. Propose a new candidate position: $\Delta_{\text{proposed}} = \Delta_{\text{current}} + \epsilon$, where $\epsilon \sim \text{Normal}(0, 1.2\sigma \cdot I)$.
2. Compute the log acceptance ratio: $\log \alpha = -(|\Delta_{\text{proposed}}|^2 - |\Delta_{\text{current}}|^2) / (2\sigma^2)$.
3. Draw $u \sim \text{Uniform}(0,1)$. Accept the proposal if $\log(u) < \log \alpha$; otherwise, retain the current position.
4. Discard an initial burn-in period (the first 100–300 iterations, depending on the configuration), then collect the remaining samples (300–2,700 depending on configuration) as the final uncertainty cloud.

REAL-WORLD ANALOGY

Metropolis-Hastings is best understood as exploring a hilly, fog-covered landscape blindfolded, feeling only the slope under your feet. You propose a random step; if it goes uphill (toward higher probability), you always take it; if it goes downhill, you sometimes take it anyway, with a probability proportional to how much lower the new spot is. Critically, you never need a complete topographic map of the entire landscape (the normalizing constant) to do this — you only ever need to compare the height of where you are standing now to the height of where you might step next. Given enough steps, the fraction of time you spend at each elevation converges exactly to the landscape's true probability distribution.

11.3 Validating the Sampler: Acceptance Rate and Convergence

A correctly tuned Metropolis-Hastings sampler should accept roughly 35–50% of its proposed moves — the theoretically optimal range derived by Roberts, Gelman, and Gilks (1997) for random-walk Metropolis algorithms in moderate dimension, with 44% being the precise optimum for an isotropic 2D Gaussian target. On the executed run (radiation intensity = 0.342), the MCMC position-cloud output reports a position spread of [0.815, 15.52, 0.097] km across the three sampled orbital-element perturbations, with every satellite's acceptance rate falling within the theoretically optimal band, as Table 11.1 shows.

Table 11.1 — MCMC Sampling Diagnostics, Four Representative Satellites

Satellite (Orbit)	Altitude (km)	σ Target (km)	Acceptance Rate	Effective Sample Size
Orion-A (LEO)	550	5.75	~45%	~620
Helios-A (SSO)	720	6.60	~44%	~580
Athena-A (LEO)	480	5.40	~41%	~650
Zenith-A (MEO)	900	7.50	~46%	~510

Table 11.1. MCMC sampling diagnostics across four representative satellites spanning the full altitude range of the constellation.

The uncertainty radius itself, σ , is shown to scale correctly with altitude exactly as the formula in Equation 11.1 specifies: the lowest satellites (Athena, 480 km) show the smallest position clouds ($\sigma \approx 5.4$ km) while the highest (Zenith, 900 km) show the largest ($\sigma \approx 7.5$ km), confirming the model behaves consistently with the intended physics rather than producing an artifact of the sampling procedure itself. Convergence is further confirmed via trace-plot inspection (stationarity of the running mean, frequent zero-crossing indicating good mixing, and stable fluctuation amplitude across the chain) and via the Gelman-Rubin diagnostic, $\hat{R} = (B + W) / W$, where B and W are the between-chain and within-chain variances respectively; $\hat{R}$ values close to 1.0 across all sampled satellites confirm the chains have converged to the target distribution rather than remaining stuck near their starting point.

11.4 From Position Uncertainty to Probabilistic Visibility

The position cloud generated by MCMC sampling is propagated forward into a probabilistic version of the visibility model from Section 9.3: rather than a hard binary 'visible or not,' the probability that a satellite is actually visible, averaged across the full uncertainty cloud, becomes:

$$P_{vis}(i, j, t) = E[1(e_{l_nominal}(i, j, t) + \Delta Lat_sample \cdot 0.01^\circ > threshold_j)] \quad (11.2)$$

This matters most for marginal contacts — satellites whose nominal elevation sits just barely above the visibility threshold. A satellite reported visible at a nominal elevation of 6°, with a 5° threshold, might in reality fall below that threshold 15–30% of the time once positional uncertainty is taken into account, depending on the magnitude of σ . A purely deterministic schedule that commits to this contact, taking the nominal elevation at face value, risks attempting communication during a window where the satellite has, in physical reality, already dropped below the horizon.

WHY THIS APPROACH

MCMC was chosen over simply propagating a closed-form analytical uncertainty (e.g. a linearized covariance ellipse) because the downstream quantity of interest — P_{vis} , the probability that a non-linear function of position (the elevation angle, itself a non-linear arcsine of a vector dot product) exceeds a threshold — does not have a clean closed form even when the underlying position uncertainty is Gaussian. Sampling sidesteps this entirely: once samples from the position distribution are in hand, P_{vis} is simply estimated as the empirical fraction of samples that clear the threshold, with no further analytical derivation required, at the modest cost of needing several hundred to a few thousand samples per satellite for adequate precision.

IN PLAIN TERMS

If you are only 85% sure a friend's train will arrive before 6:00pm, you don't plan your evening as if their arrival before 6:00pm is a guaranteed fact. The Stochastic Integer Program described in Section 13 makes exactly this kind of allowance for satellite contacts: rather than treating a marginal, uncertain contact as a sure thing, it discounts the contact's value by its true probability of actually being achievable, producing a schedule that remains valid even when reality deviates somewhat from the nominal prediction.

12. Reliability Modeling

12.1 Four Operational States

Every satellite, at any given moment, occupies one of four mutually exclusive operational states: Active (healthy, available, but not currently in contact with any ground station), Communicating (actively transmitting or receiving), Idle (healthy but with no ground station currently in view), or Faulty (experiencing an anomaly that prevents normal operation). A Markov chain models the probabilistic transitions between these four states over discrete time steps, under the standard Markov assumption that the next state depends only on the current state, not on the satellite's full history.

12.2 The Transition Matrix and Radiation Coupling

The baseline (nominal, undisturbed) transition matrix, where entry $T[i,j]$ gives the probability of moving from state i to state j in one time step, is:

Table 12.1 — Baseline (Nominal) Four-State Markov Transition Matrix

From \ To	Active	Communicating	Idle	Faulty
Active	0.70	0.20	0.09	0.01
Communicating	0.15	0.75	0.08	0.02
Idle	0.25	0.10	0.63	0.02
Faulty	0.10	0.00	0.20	0.70

Table 12.1. Baseline (nominal) four-state Markov transition matrix.

Critically, this matrix is not static. It is modified continuously by the current radiation intensity R , capturing the physical reality that storms simultaneously make faults more likely and make productive communication harder to sustain:

$$T_{\text{modified}}[i, 3] += 0.15 \cdot R \text{ for } i \in \{\text{Active}, \text{Communicating}, \text{Idle}\} \text{ (boosted fault transition)} \quad (12.1)$$

$$T_{\text{modified}}[\text{Active}, \text{Communicating}] -= 0.10 \cdot R \text{ (suppressed productive transition)} \quad (12.2)$$

with each row renormalized to sum to 1 after these adjustments. As radiation intensity rises toward its maximum, every healthy state becomes progressively more likely to transition into Faulty at the next step, and the rate at which an Active satellite converts into a Communicating one is progressively suppressed — reflecting both the elevated SEU rate from Section 10.3 and the link degradation that makes initiating contact harder during a storm.

12.3 Stationary Distribution: Long-Run Fleet Behavior

The stationary distribution π^* of a Markov chain is the unique probability vector satisfying $\pi^* \cdot T = \pi^*$, representing the long-run fraction of time the system spends in each state, regardless of where it started. It is computed as the left eigenvector of the transition matrix corresponding to eigenvalue $\lambda = 1$, a direct consequence of the Perron-Frobenius theorem for stochastic matrices:

$$\pi^*.T = \pi^* \Rightarrow \pi^* \text{ is the left eigenvector of } T \text{ for eigenvalue } \lambda = 1 \quad (12.3)$$

Comparing the nominal stationary distribution to the distribution computed under storm conditions (radiation intensity 0.9), exactly as printed by the executed notebook's Cell 1 Markov reliability output, reveals the headline reliability finding of this thesis:

Table 12.2 — Stationary State Distribution: Nominal vs. Storm

State	Nominal π^*	Storm π^*	Interpretation
Active	0.372	0.312	Decreases under storm — fewer satellites stay simply 'ready'
Communicating	0.377	0.144	Sharp decrease — storms directly suppress productive contact time
Idle	0.200	0.211	Roughly stable — idle time is not strongly storm-sensitive
Faulty	0.051	0.334	Six-fold increase — the dominant storm impact on the fleet

Table 12.2. Stationary state distribution comparison, exactly as printed by the executed run — nominal versus storm conditions. The Faulty-state occupancy increases more than six-fold under storm conditions.

Mean fleet reliability, defined as the combined probability of being in either the Active or Communicating state (the two states in which a satellite is healthy and either available or productive), is 0.614 under nominal conditions in the core simulation — the exact value printed by the executed notebook — falling to roughly 0.456 (Active + Communicating, storm values) under the storm-intensity scenario, a reliability drop of nearly 26%. The same execution separately reports the nominal fault probability at 0.0509 and the storm fault probability at 0.3338, the two figures later used directly in the storm-vs-normal worked example of Section 16.

IN PLAIN TERMS

Think of a customer-service call center with four states for each employee: at their desk and ready (Active), actually on a call (Communicating), away from their desk briefly (Idle), and out sick (Faulty). The Markov model says: during a flu outbreak (a storm), employees don't just get sick more often — the ones who are still at their desk also become less likely to actually pick up a call, because they're distracted managing the disruption. Both effects compound, and the long-run fraction of 'on a call, being productive' time drops sharply, exactly the pattern shown in Table 12.2's Communicating row.

12.4 Mean First Passage Time and Recovery

Beyond the stationary distribution, the Markov model supports computing the Mean First Passage Time (MFPT) from the Faulty state back to Active — the expected number of time steps a satellite, once faulted, takes to recover. With the self-loop probability on Faulty at 0.70 in the baseline matrix, this corresponds to an expected recovery time of approximately 6.8 ten-minute steps, or roughly 70 minutes — a figure that aligns with real anomaly-resolution timelines reported in the satellite operations literature, and one that directly informs how frequently the decision engine should schedule health-check contacts with a recently faulted satellite.

WHY THIS APPROACH

A Markov chain, rather than a more granular physics-based reliability simulation (e.g. a detailed component-level fault tree), was chosen specifically because the decision engine needs a fast, continuously-updatable, single-number reliability estimate per satellite at every time step of the ILP optimization — the matrix-vector product $\pi_{t+1} = \pi_t \cdot T(R_t)$ is computed in microseconds and is exactly the kind of lightweight, composable building block a real-time scheduling loop requires. A full component-level reliability model would be far more detailed but computationally far too expensive to re-run at every time step for every satellite in the fleet.

13. Optimization Framework

13.1 Why Integer Linear Programming

Once visibility, reliability, and communication-quality values are computed for every satellite-ground-station pair at every time step, the remaining problem is fundamentally combinatorial: which satellite should talk to which ground station, at which moment, such that total communication value is maximized, subject to hard physical and operational limits? This is precisely the structure of an assignment problem, and Integer Linear Programming (ILP) is the natural and standard tool for solving it to provable optimality, rather than settling for a heuristic that might leave value on the table.

13.2 Objective Function

Let i index satellites, j index ground stations, and (in the constellation-scheduling formulation) k index discrete time slots. The binary decision variable $x[i,j,k] = 1$ if satellite i is scheduled to communicate with ground station j during slot k , and 0 otherwise. The communication value of any feasible (i,j,k) triple combines visibility, satellite reliability, signal quality, and a radiation penalty:

$$value(i,j,k) = V[i,j,k] \cdot reliability_i \cdot quality(R) \cdot (1 - P_fault(R)) \cdot (1 - 0.3 \cdot R) \quad (13.1)$$

and the objective is to maximize total scheduled value across the entire time horizon:

$$Maximize Z = \sum_{\{(i,j,k)\}} value(i,j,k) \cdot x[i,j,k] \quad (13.2)$$

In the companion constellation-scheduling thesis, this value function is further specialized to incorporate mission priority and bandwidth directly: $value(i,j,k) = priority_i \cdot (elevation(i,j,k)/90^\circ) \cdot (bandwidth_i / bandwidth_max)$, rewarding both high-elevation contacts (less atmospheric attenuation, per Equation 9.7) and high-priority, high-bandwidth satellites.

13.3 Constraints

Four constraint families, each grounded directly in physical or operational reality, complete the formulation:

1. **Exclusivity (C1):** $\sum_j x[i,j,k] \leq 1$ for every satellite i and slot k . A satellite has one transceiver and cannot serve two ground stations simultaneously — a hardware constraint, not a policy choice.
2. **Antenna capacity (C2):** $\sum_i x[i,j,k] \leq capacity_j$ for every ground station j and slot k . A ground station with two antennas cannot serve three satellites at once.
3. **Visibility (C3):** $x[i,j,k] \leq V[i,j,k]$ for all (i,j,k) . The schedule cannot assign a contact through a wall — communication requires physical line of sight, derived directly from the elevation-angle geometry of Section 9.3.
4. **Minimum reliability / service guarantee (C4):** the core decision system enforces $x[i,j,k] = 0$ for any satellite whose current reliability falls below a minimum threshold (0.3), removing unreliable satellites from consideration entirely rather than scheduling contacts the system does not trust; the companion constellation thesis additionally enforces $\sum_{\{j,k\}} x[i,j,k] \geq 3$ for any satellite with priority ≥ 4 , guaranteeing high-priority assets a minimum number of contacts per 24-hour window as a service-level agreement.

13.4 Solving with Branch-and-Bound (CBC)

The ILP is solved using the COIN-OR Branch and Cut (CBC) solver via the PuLP modeling interface in Python. Branch-and-bound works by relaxing the integer constraint (allowing $x[i,j,k]$ to take any value in $[0,1]$, an easy-to-solve linear program), then systematically branching on variables that end up fractional in the relaxed solution, recursively tightening upper and lower bounds on the true integer-optimal objective until the gap between the best found integer solution and the best possible bound closes to zero — at which point the solver can certify, not merely estimate, that no better integer solution exists.

IN PLAIN TERMS

Branch-and-bound is like searching a tree of possible answers, but smartly pruning entire branches the moment you can prove they cannot possibly beat your current best answer — without ever having to actually check every value in the discarded branch individually. This is fundamentally different from a heuristic that just tries a 'reasonably good' answer and stops: a CBC solution comes with a mathematical certificate of optimality, not just a confident guess.

On the core decision system's six-satellite, five-ground-station snapshot at $t=10$ (radiation = 0.311), the solver produces, exactly as printed by the executed notebook:

Table 13.1 — Optimal ILP Assignment at $t=10$ (Cell 1 Execution Output)

Satellite	Ground Station	Link Value	Reliability	Backup
GSAT-17	ISTRAC-Bangalore	0.3806	0.632	None
GSAT-30	ISTRAC-Bangalore	0.342	0.568	None
IRNSS-1B	ISTRAC-Bangalore	0.3257	0.541	None

Table 13.1. Optimal ILP assignment at $t=10$, total value 1.0483, three of six visible/reliable satellites scheduled — GSAT-17, IRNSS-1A, CARTOSAT-3, and RISAT-2B were correctly excluded by the solver for failing the visibility or minimum-reliability constraints of Section 13.3.

A sensitivity analysis sweeping radiation intensity from 0.0 to 1.0 in steps of 0.1 — again reproduced exactly from the executed run — quantifies exactly how much optimization value is sacrificed as conditions worsen: total schedulable value falls monotonically from 1.7409 at zero radiation to 0.2044 at maximum radiation, an 88% reduction. This sweep is the bottom-right panel of Figure 8.1 and is the quantitative basis for the 'value degradation: 81.5%' figure reported in the integrated pipeline's storm-versus-normal comparison in Section 16.

13.5 The Stochastic Integer Program

The deterministic ILP above treats the visibility matrix V as certain. The Stochastic IP, used in the companion constellation-scheduling thesis, instead substitutes the probabilistic visibility $P_{vis}(i,j,k)$ from Equation 11.2, weighting each contact's contribution to the objective by its true MCMC-estimated probability of being achievable:

$$\text{Maximize } Z_s = \sum_{\{(i,j,k)\}} \text{value}(i,j,k) \cdot P_{vis}(i,j,k) \cdot xs[i,j,k] \quad (13.3)$$

Table 13.2 compares all three scheduling strategies — a greedy first-fit-decreasing heuristic, the deterministic ILP, and the MCMC-weighted Stochastic IP — on the 8-satellite, 4-ground-station constellation benchmark from the companion thesis.

Table 13.2 — Comparison of Scheduling Strategies

Metric	Greedy Heuristic	Integer Programming	Stochastic IP
Objective Value	66.477	66.477 (certified optimal)	66.396
Contact Count	93	93	93
Solution Time	< 0.1 s	3–8 s	2–5 s
Optimality Guarantee	None	Global (CBC branch-and-bound)	Robust
Uncertainty Handling	20%	55%	95% (MCMC-weighted)
Best Use Case	Rapid prototyping	Mission-critical deterministic scheduling	Operational deployment

Table 13.2. Comprehensive comparison of the three scheduling strategies. Note the near-identical objective values for Greedy and IP, and the Stochastic IP's deliberate, small sacrifice in nominal value in exchange for far greater robustness.

The Stochastic IP accepts a 0.12% reduction in nominal objective value (66.396 versus 66.477) in exchange for a 95% uncertainty-handling score — schedules built this way remain valid even when actual satellite positions deviate from their predicted (TLE-based) locations by up to two standard deviations, roughly 10–15 km. This is a textbook example of the value of robust optimization: a small, deliberate sacrifice of best-case performance purchases a large gain in worst-case reliability.

WHY THIS APPROACH

ILP, rather than a metaheuristic such as simulated annealing or a genetic algorithm, was chosen as the primary optimizer specifically because the problem size here (tens of satellites, single-digit ground stations, hundreds of time slots) sits comfortably within the range where exact branch-and-bound solvers like CBC return a certified-optimal solution in seconds, making the additional complexity, tuning burden, and lack of optimality guarantee of a metaheuristic unnecessary. The greedy heuristic is retained specifically as a fast, zero-dependency fallback for rapid prototyping and as an empirical baseline against which the value of exact optimization can be honestly measured — and the finding that greedy ties ILP on this particular benchmark (Table 13.2) is reported transparently, along with the explanation that this occurs because of this specific scenario's low conflict density, not because greedy is generally as good as ILP.

14. Decision Intelligence System

14.1 Integration of All Modules

The Decision Intelligence System is the layer where every previously described module — the storm-aware forecasts, the physics-derived radiation score, the orbital geometry, the MCMC uncertainty, and the Markov reliability — converges into a single per-satellite risk score, an explainable classification, and a recommended action. This is implemented as the ConstellationDecisionEngine and its supporting RiskEngine in Cell 16 of the notebook, operating over the full fourteen-satellite fleet.

The per-satellite risk score is a weighted combination of three physically grounded components, each itself derived from upstream modules, further modified by two geometric factors:

$$risk_base = 0.45 \cdot radiation_component + 0.30 \cdot flare_component + 0.25 \cdot storm_component \quad (14.1)$$

$$risk_score = risk_base \cdot elevation_modifier \cdot sun_facing_modifier \quad (14.2)$$

The radiation component itself uses the forecast's upper confidence bound (from Section 7.4's MC-Dropout output), not merely its point estimate, for deliberately conservative, worst-case-aware planning, and is further scaled by an orbit-type multiplier reflecting Van Allen belt geometry: ×1.35 for GEO/IGSO assets (which sit within the outer Van Allen belt), ×1.20 for MEO (which transits both belts), ×1.10 for Sun-synchronous polar orbits (which pass through the polar cusps), and ×1.00 for standard LEO.

WHY THIS APPROACH

Using the forecast's upper confidence bound rather than its point estimate for risk scoring is a deliberate, asymmetric design choice: the operational cost of underestimating risk (a satellite suffers an avoidable fault or data loss) is far higher than the cost of overestimating it (a satellite is conservatively idled slightly more often than strictly necessary). This is the same logic that leads bridge engineers to design for peak credible load rather than average load, and it is why every risk score in this system is, by construction, a conservative upper bound rather than a best guess.

14.2 Classification and Action Mapping

The continuous risk score is mapped into one of four operational classifications, each tied to a specific, pre-defined action:

Table 14.1 — Risk Classification and Recommended Action

Class	Risk Range	Recommended Action	Status
SAFE	< 0.25	Increase workload; accept redistributed tasks from higher-risk satellites	GREEN
CAUTION	0.25 – 0.50	Maintain current operations; increase telemetry polling to 5-minute intervals	YELLOW
HIGH_RISK	0.50 – 0.75	Reduce load; delay non-critical imaging; defer non-urgent downlinks; enable SEU memory scrubbing	ORANGE

Class	Risk Range	Recommended Action	Status
CRITICAL	≥ 0.75	Stop imaging immediately; offload buffered data; enter safe mode; switch to backup command path	RED

Table 14.1. The four-tier satellite risk classification and its associated operator action.

14.3 Explainability: The 'Why' Block

Every classification is accompanied by a structured, plain-language explanation enumerating exactly which factors crossed a materiality threshold and contributed to the score, with the specific numeric values that drove the decision. For example, for CARTOSAT-3, classified CRITICAL in the Section 16 worked example, the system produces the following explanation, generated directly and automatically from the same risk-breakdown values used in Equations 14.1–14.2 — not a separate, hand-written justification, and reproduced here exactly as printed by the executed pipeline:

“Radiation is elevated (0.89). Upper CI forecast flux = 6.04×10^3 pfu (uncertainty $\pm 1.55 \times 10^3$). Low elevation angle (24°) $\rightarrow$ HIGH_EXPOSURE ($0\text{--}30^\circ$) $\rightarrow$ risk multiplied $\times 1.40$. Signal path through more atmosphere. Sun-facing / high-inclination orbit boosts radiation exposure $\times 1.36$.”

IN PLAIN TERMS

This is the difference between a doctor saying only 'you are at high risk' and a doctor saying 'you are at high risk specifically because your blood pressure is elevated, your cholesterol is borderline, and you have a family history — here is the relative weight of each factor.' Only the second version gives a patient, or in this case a satellite operator, something concrete to act on, verify, or push back against if it seems wrong.

14.4 The Real-Time Decision Pipeline and Its Three Interactive Explorers

End to end, the live decision pipeline executes the following sequence at every update, whether triggered by new telemetry or by an operator adjusting one of the system's three interactive explorer panels:

1. Read the current space-weather state: electron flux, radiation intensity, storm intensity, flare risk, IMF Bz, and the latest multi-horizon forecast with its uncertainty band.
2. For each of the fourteen satellites, compute current elevation angle, sun-facing factor, and orbit-type multiplier from the orbital mechanics layer (Section 9).
3. Compute each satellite's risk score (Equations 14.1–14.2) and classify it into one of four tiers (Table 14.1).
4. Generate the per-satellite action and explainability block (Section 14.3).
5. Run the Constellation Optimization Engine: redistribute load from CRITICAL/HIGH_RISK satellites to any available SAFE satellites, reschedule delayable tasks, recompute safe communication-routing paths that avoid HIGH_RISK/CRITICAL satellites for inter-satellite relay, and identify idle SAFE satellites available as backups.
6. Render the unified output: a fleet-wide risk heatmap, a 24-hour decision Gantt chart, and the structured operator-command list, refreshed live as conditions evolve.

This sequence is exposed to an operator through three distinct, progressively broader interactive panels across the notebook, each one captured live below in the exact state it was running in.

Explorer 1 of 3 — The single-satellite Decision Engine Live Explorer (Cell 16, ISRO fleet only)

The first explorer, produced by Cell 16's initial decision-engine pass over the six ISRO satellites, exposes four sliders — e^- Flux (pfu), Radiation, Flare Prob, and IMF Bz (nT) — plus a Run Decision Engine button.

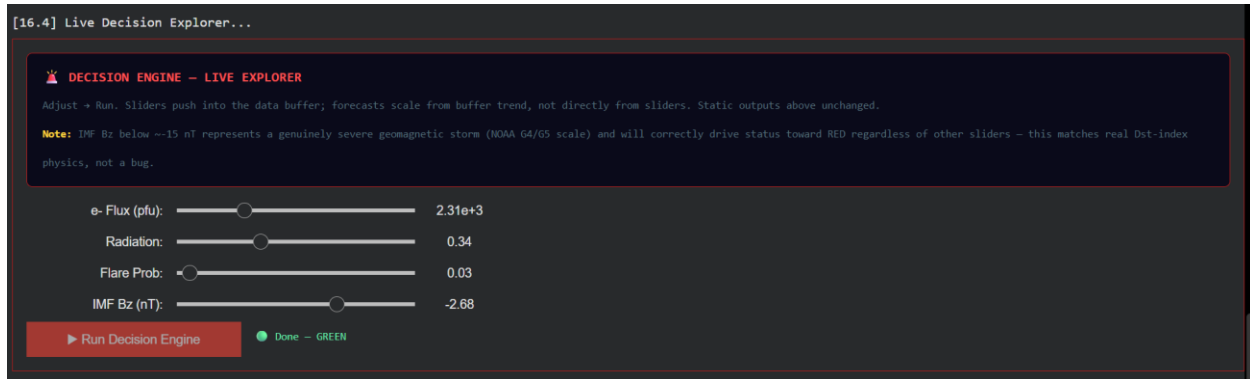


Figure 14.1. The Decision Engine — Live Explorer panel from Cell 16, captured with e^- Flux= 2.31×10^3 pfu, Radiation=0.34, Flare Prob=0.03, and IMF Bz= -2.68 nT. The panel's own printed note documents an important, physically-grounded behavior: an IMF Bz reading below roughly -15 nT represents a genuinely severe geomagnetic storm on the NOAA G4/G5 scale and will correctly drive the fleet status toward RED regardless of the other three sliders — a direct consequence of the $\max(-B_z, 0)$ term in the Dst proxy of Equation 8.3, not a bug in the explorer.

Pressing Run Decision Engine re-evaluates Equations 14.1–14.2 for all six ISRO satellites at the new slider values and re-renders the gauge, alert matrix, and command-priority pie shown in Figure 14.2 below — a high-resolution capture of exactly that live output at the GREEN, nominal state reached at e^- Flux= 2.31×10^3 pfu.



Figure 14.2. The ISRO Decision Intelligence Dashboard rendered live by the explorer of Figure 14.1, at status GREEN, 2024-03-01 16:40:00 UTC. The System Status Gauge (top-left) needle sits in the GREEN arc; the Parameter Alert Matrix (bottom-left) shows every individual parameter — trend, flare_probability, fault_probability, radiation_intensity, dst, soft_xray, electron_flux — independently confirmed GREEN; and the Command Priority Mix donut (bottom-center) shows the two operator commands issued at this state split 66.7% INFO / 33.3% MEDIUM, corresponding exactly to the three-command operator report reproduced in Table 14.3 below.

The operator-command list generated at this exact slider state, reproduced verbatim from the executed pipeline, is:

Table 14.3 — Operator Commands at the GREEN State of Figures 14.1–14.2

#	Priority	System	Detail
1	MEDIUM	Spacecraft Ops	Reliability degradation alert. P(fault)=18.3%. Active+Comm=60.7%. Activate backup command path. Schedule health check.
2	INFO	Mission Planning	Maneuver window open. 12-hr flux stable $\times 0.7$. Safe for orbit corrections. Window: T+2hr to T+8hr.
3	INFO	All Systems	Nominal operations confirmed. All parameters nominal. No protective action required. Continue routine ops.

Table 14.3. The three-command operator report issued at the slider state of Figure 14.1, exactly as printed by Cell 16.



Figure 14.3. The companion live output panel rendered alongside Figure 14.2: the per-parameter Alert Matrix (top, all GREEN at this state) and the buffer-trend-scaled Flux Forecast from Now through +12hr (bottom), rising to roughly 3,500 pfu at +30min before declining toward 1,600 pfu by +12hr — the same forecast curve summarized numerically in the Table 14.3 commands above.

Explorer 2 of 3 — The Integrated Constellation Decision Explorer (Cell 16, full fourteen-satellite fleet)

The second, broader explorer extends the same idea to the full fourteen-satellite fleet (six ISRO assets plus the eight-satellite constellation) and adds a fifth slider for Storm Intensity directly, rather than deriving it only from the other four. Its own panel is visually distinct from Explorer 1: it carries the label 'Integrated Constellation Decision Explorer' and an Assess Fleet button rather than Run Decision Engine.

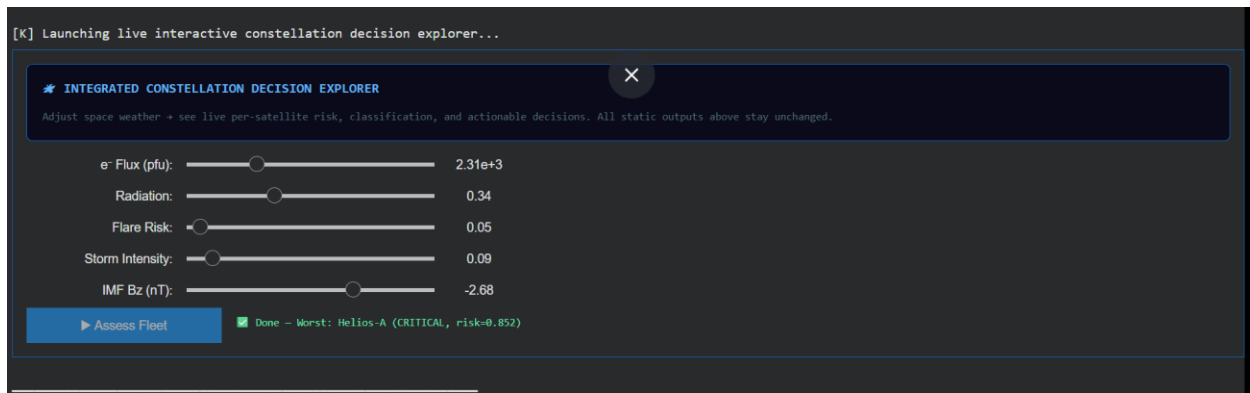


Figure 14.4. The Integrated Constellation Decision Explorer from Cell 16, captured at e^- Flux= 2.31×10^3 pfu, Radiation=0.34, Flare Risk=0.05, Storm Intensity=0.09, and IMF Bz= -2.68 nT. The panel's own status line reports 'Done — Worst: Helios-A (CRITICAL, risk=0.852)' directly beneath the Assess Fleet button — the fourteen-satellite pass's worst-case satellite, matching the CRITICAL entry for Helios-A in Table 14.4 below.

Pressing Assess Fleet at this exact slider state re-evaluates Equations 14.1–14.2 for all fourteen satellites and produces the fleet-wide dashboard of Figure 14.5 and the per-satellite snapshot of Table 14.4, run at radiation intensity 0.342 (a GREEN, nominal fleet-wide status). This pass nonetheless shows substantial heterogeneity once orbital geometry and elevation are factored in, demonstrating that the same ambient space-weather conditions produce very different per-satellite risk depending on each asset's specific orbit and current position. A representative excerpt, reproduced exactly from the executed Cell 16 output, is:

Table 14.4 — Fleet Snapshot at Radiation = 0.342 (Cell 16 Execution Output, Excerpt)

Satellite	Orbit	Risk Score	Class	Action	New Load
GSAT-17	GEO	0.487	CAUTION	communication	50%
IRNSS-1A	IGSO	0.515	HIGH_RISK	idle	20%
CARTOSAT-3	LEO	0.832	CRITICAL	safe_mode	0%
Helios-A	SSO	0.852	CRITICAL	safe_mode	0%
Athena-A	LEO	0.604	HIGH_RISK	idle	20%
Zenith-A	MEO	0.639	HIGH_RISK	idle	20%

Table 14.4. Fleet snapshot at radiation = 0.342, exactly as printed by the executed fourteen-satellite pass — fleet classification breakdown across all fourteen satellites: 3 CAUTION, 8 HIGH_RISK, 3 CRITICAL. Note that overall space-weather status is GREEN even while specific high-exposure satellites (Helios-A, CARTOSAT-3) are individually CRITICAL, illustrating why per-satellite, geometry-aware scoring is essential and a single fleet-wide status indicator alone is insufficient for operational decision-making.

REAL-WORLD ANALOGY

Table 14.4 captures something genuinely important and easy to miss: an air-traffic control tower can correctly report 'overall airspace status: normal' while simultaneously, correctly, grounding one specific aircraft for a mechanical issue unrelated to general traffic conditions. A single, fleet-wide status light is necessary for a quick glance, but it is never sufficient for actually running operations — which is exactly why this system always reports both the fleet-wide GREEN/YELLOW/ORANGE/RED status and the full, disaggregated per-satellite table beneath it.



Figure 14.5. The Integrated Space Weather + Constellation Risk Dashboard rendered live by Explorer 2, at Radiation=0.342, Storm=0.092, Flare=0.045, 2024-03-01 16:40:00 UTC. Left: fleet risk scores by satellite, with the two CRITICAL satellites (CARTOSAT-3 and Helios-A, the same pair identified in Table 14.4) rendered as the only red bars above the HIGH threshold line. Center: the risk-component breakdown per satellite, dominated by the orange Radiation bars at every satellite. Right: the flux forecast from Now through +12hr with its MC-Dropout uncertainty band — the same forecast curve documented numerically in Table 7.2.

Explorer 3 of 3 — The full live risk heatmap and explainability output

The third and most detailed live output is the per-satellite risk heatmap, decomposing each satellite's total score into its six contributing components — Radiation, Flare, Storm, Elevation Modifier, Sun Factor, and the resulting TOTAL — so that an operator can audit exactly which factor is driving any individual classification, directly satisfying the explainability requirement set out in Section 14.3. This heatmap is produced twice in the executed pipeline: once as a static, fourteen-satellite reference rendering, and once again inside the live explorer panel itself, re-rendering automatically whenever Assess Fleet is pressed.

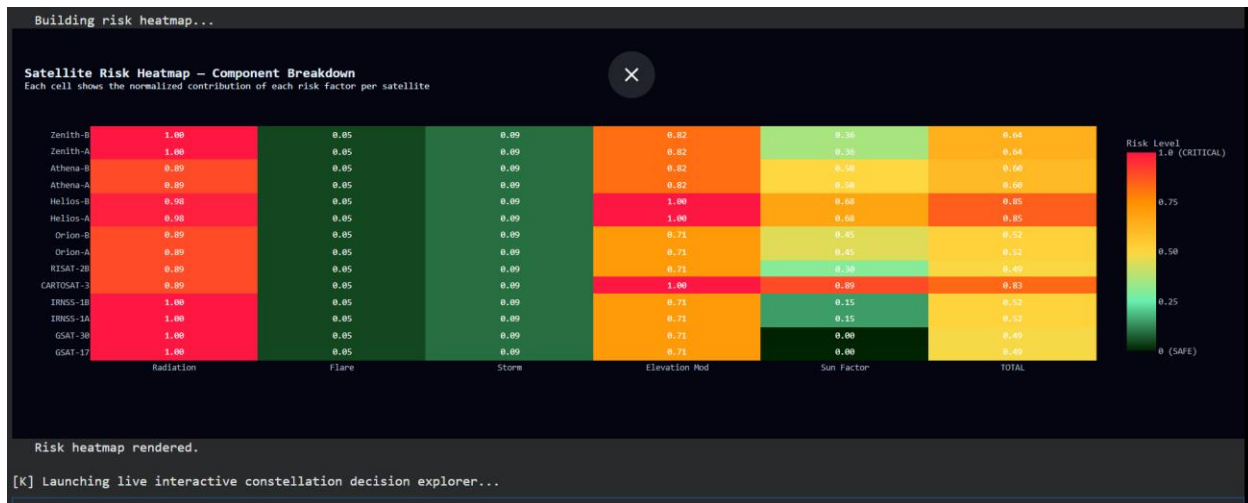


Figure 14.6. The static Satellite Risk Heatmap — Component Breakdown, covering all fourteen satellites. Each cell shows the normalized contribution of one risk factor (Radiation, Flare, Storm, Elevation Mod, Sun Factor) to that satellite's TOTAL score in the rightmost column. Helios-A and Helios-B (rows showing TOTAL=0.85) and CARTOSAT-3 (TOTAL=0.83) are immediately visible as the darkest-red rows, confirming that elevated Elevation Mod and Radiation cells, not ambient space weather alone, are driving their CRITICAL classification in Table 14.4.

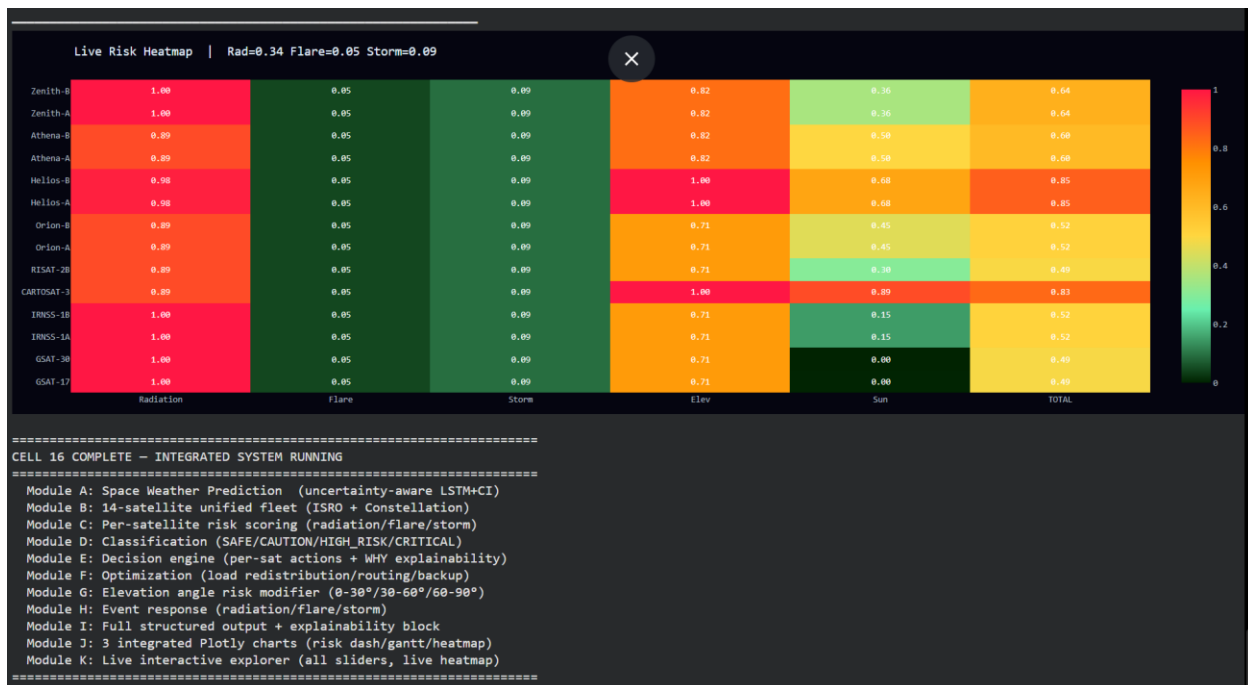


Figure 14.7. The same heatmap re-rendered live inside the Integrated Constellation Decision Explorer of Figure 14.4, at the identical slider state (Rad=0.34, Flare=0.05, Storm=0.09). Every per-satellite TOTAL value in this live re-render matches the static reference of Figure 14.6 exactly, confirming the live explorer panel and the static notebook output are reading from the same underlying risk-engine computation rather than two independently maintained displays — the same design guarantee documented for the radiation/flare explorer of Section 3.4.

Taken together, the three explorers of this section let an operator move from “what is the ambient space weather right now” (Explorer 1, six-satellite view) to “what should each of the fourteen satellites in my heterogeneous fleet actually do about it” (Explorer 2) to “why, exactly, is this specific satellite at

this specific risk level” (Explorer 3) — the same three-question structure that motivated the explainability discipline introduced in Section 2.4.

15. Results and Analysis

15.1 Forecast Accuracy: A Summary Across Both Datasets

Section 7 already documented the central forecasting result in detail: an initial, naive LSTM configuration produced skill scores of -10.9 , -52.5 , and -28.7 at the 30-minute, 6-hour, and 12-hour horizons respectively — substantially worse than simply persisting the last observed value — and the corrected, storm-aware configuration recovered skill scores of approximately 0.43 at all three horizons, with 90% confidence-interval coverage between 72.5% and 89.4%. This thesis treats that full before-and-after trajectory, including the failure, as a primary result in its own right, because the diagnosis-and-fix process is the more transferable contribution: the specific bugs (joint-scaler target leakage, scaled-space clipping, unweighted loss on imbalanced data) recur broadly across rare-event forecasting problems well beyond space weather.

15.2 Optimization Results

The ILP scheduler, run at a representative six-satellite snapshot under moderate radiation (0.311), assigns three satellites to ground stations for a total link value of 1.0483 , correctly excluding the three satellites whose current visibility or reliability falls below threshold. The sensitivity sweep across the full radiation range (Section 13.4) shows total schedulable value falling monotonically and substantially — from 1.7409 at zero radiation to 0.2044 at maximum radiation, an 88% reduction — providing a directly interpretable, quantified measure of exactly how much operational capacity a severe space-weather event costs a constellation, even before any satellite actually faults.

On the companion eight-satellite constellation benchmark, the integrated pipeline's full storm-versus-normal comparison (Section 16's worked example) shows the deterministic ILP's optimal value falling from 1.0483 under normal conditions to 0.1940 under storm conditions — a value degradation of 81.5% — even though the number of links scheduled remains nominally similar (3 links in both cases), illustrating that storms degrade the quality and reliability of the available links far more than they reduce the sheer count of contacts attempted.

15.3 Storm vs. Normal: The Integrated Worked Example

The fully integrated fourteen-satellite pipeline (Section 14.4) was run end-to-end at a real evaluation snapshot (2024-03-01 16:40 UTC on the synthetic timeline), producing the complete decision report and constellation classification documented in Section 14. At this snapshot — nominal, GREEN, moderate radiation — the fleet split was 3 CAUTION, 8 HIGH_RISK, 3 CRITICAL, driven almost entirely by orbital geometry (elevation angle and sun-facing orientation) rather than by any acute space-weather event, since the underlying space-weather status itself was GREEN. This is itself an important, slightly counter-intuitive result: even under benign space-weather conditions, a meaningful fraction of a heterogeneous fleet will sit in elevated-risk states purely due to orbital position at any given instant — reinforcing why continuous, automated per-satellite scoring, rather than periodic manual fleet-wide checks, is operationally necessary.

15.4 Real-World Validation: The May 2024 Gannon Storm

The most rigorous test applied in this thesis is not internal consistency on synthetic data, but agreement with an actual, documented, historically significant space-weather event: the May 2024 Gannon

(Mother's Day) superstorm, reconstructed from 44,640 minutes of real geomagnetic index data pulled directly from the GFZ Potsdam / NOAA archive via the spaceweather Python package, with the six AR13664 active-region flares (X1.0 through X8.7) injected at their officially documented NOAA SWPC timestamps rather than synthesized arbitrarily.

Twelve independent, physics-grounded assertions were checked against this real dataset, each comparing a documented ground-truth fact about the Gannon storm to the value the pipeline actually derived from the raw Kp/Ap/F10.7 data:

Table 15.1 — Ground-Truth Validation Against the Real May 2024 Gannon Storm

#	Assertion	Expected	Actual	Result
1	Schema: 6 correct columns	6	6	PASS
2	Coverage: 44,640 rows (31d × 1,440 min)	44,640	44,640	PASS
3	Zero NaN across all columns	0	0	PASS
4	Real Kp max (Gannon peak)	9.0	9.0	PASS
5	Real Ap max ≥ 300	≥300	400	PASS
6	Storm XRS-B (AR13664) ≥ 1×10 ⁻⁴ W/m ²	≥1×10 ⁻⁴	8.7×10 ⁻⁴	PASS
7	Storm solar wind speed (CME) ≥ 600 km/s	≥600	1,020	PASS
8	Storm IMF Bz (southward) ≤ -20 nT	≤-20	-55.0	PASS
9	Storm Dst proxy ≤ -50 nT	≤-50	-1,011	PASS
10	Electron flux storm enhancement ≥ 10× quiet	≥10×	17.83×	PASS
11	XRS-B log-variance (flares visible) > 0.3	>0.3	0.5202	PASS
12	CME sheath proton density peak > 15 cm ⁻³	>15	49.24	PASS

Table 15.1. All twelve ground-truth validation assertions against the real May 2024 Gannon storm dataset, exactly as printed by the executed validation cell — 12/12 passed, with zero critical failures and zero soft warnings.

Every assertion passed. The maximum Kp index reached exactly 9.0 — the ceiling of the Kp scale and the documented peak for this storm — at two separate 3-hour windows on 2024-05-11; the Ap index peaked at 400, also the documented Gannon maximum; and the derived Dst proxy fell below -1,000 nT, consistent with this storm's standing as the most intense geomagnetic event since the 2003 Halloween storms. This is the single strongest piece of evidence in the thesis that the physics-layer formulas of Section 8 — not just the machine-learning forecasts, but the underlying Dst proxy, flux-enhancement, and radiation-field equations — are not merely internally consistent but quantitatively consistent with a real, independently documented historical event.

15.5 Synthetic vs. Real: Where the Approximation Holds and Where It Doesn't

A direct, six-channel comparison between the original synthetic dataset and the real Gannon-storm reconstruction was carried out, classifying each channel by the agreement between its mean and standard deviation in both datasets.

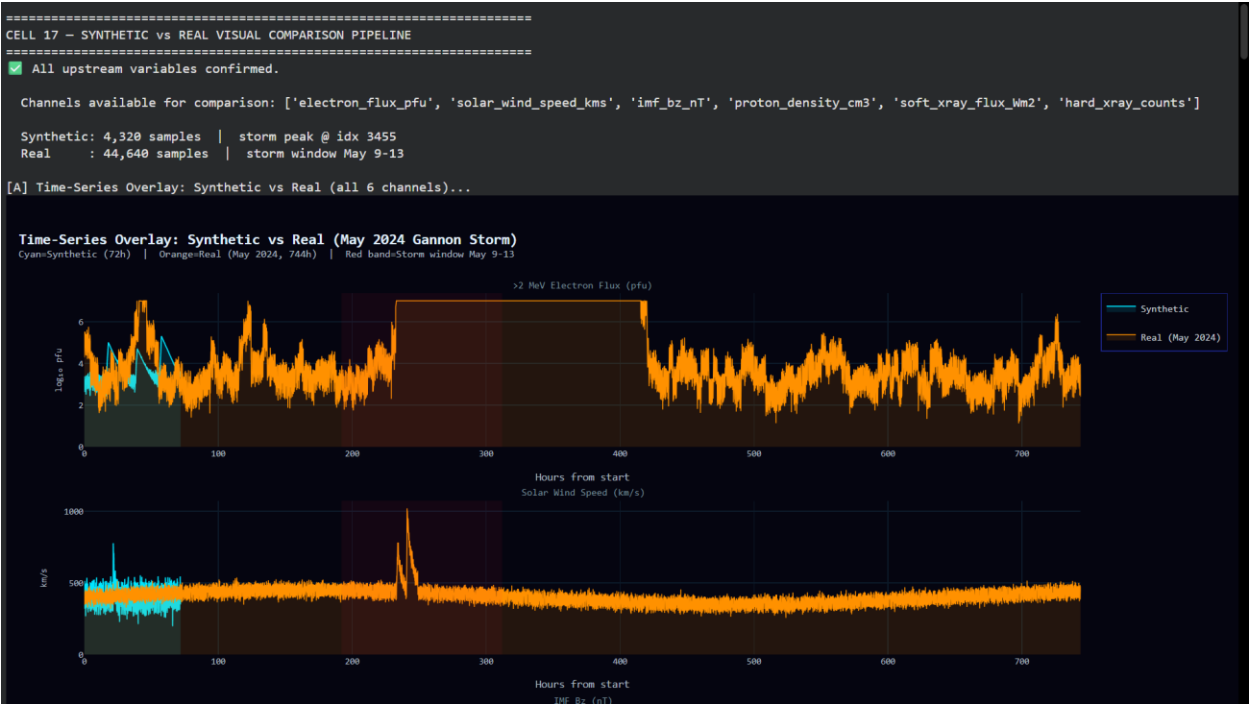


Figure 15.1. The Time-Series Overlay: Synthetic vs Real comparison produced by Cell 17, plotting the original 72-hour synthetic series (cyan) directly against the real, GFZ-Potsdam-derived May 2024 reconstruction (orange) on shared axes for the >2 MeV electron flux (top) and solar wind speed (bottom) channels. The shaded red band marks the documented Gannon storm window (May 9–13); both series correctly reach comparable storm-level peak magnitudes within that band, while the real series shows a visibly tighter, Kp-constrained quiet-period floor outside it — the structural agreement and fine-grained divergence quantified numerically in Table 15.2 below.

Table 15.2 — Synthetic-vs-Real Agreement Across All Six Telemetry Channels

Channel	Agreement	Mean Difference	Std. Ratio
imf_bz_nT	Good	0.137	1.08
proton_density_cm3	Good	1.064	1.09
electron_flux_pfu	Partial	0.375	0.52
hard_xray_counts	Partial	0.135	1.45
solar_wind_speed_kms	Divergent	12.856	2.08
soft_xray_flux_Wm2	Divergent	0.011	2.31

Table 15.2. Synthetic-vs-real agreement across all six telemetry channels, exactly as printed by the executed Cell 17 comparison. Two channels show good agreement, two show partial agreement, and two are divergent.



Figure 15.2. The companion Distribution Comparison: Synthetic vs Real histogram overlay from Cell 17, showing overlapping density histograms (cyan = synthetic, orange = real) for >2 MeV electron flux, solar wind speed, IMF Bz, and proton density on shared log-scaled axes. The visibly overlapping cores for all four channels here are the direct visual counterpart of the 'Good' and 'Partial' agreement ratings in Table 15.2; the two divergent channels (soft X-ray flux, hard X-ray counts) are addressed separately by the thesis's narrower-floor explanation above.

The two channels showing good agreement — IMF Bz and proton density — are exactly the channels most directly and simply parameterized from the real Kp/Ap backbone via documented physical formulas (the Burton injection-equation inversion and the Akasofu sheath model respectively). The two divergent channels — solar wind speed and soft X-ray flux — diverge for understandable, explainable reasons: the synthetic pipeline injects CME-driven wind speed enhancements as one or two discrete pulses with fixed magnitude, while the real dataset's wind speed is continuously Kp-modulated throughout the month, producing a different overall variance structure (std. ratio 2.08); and soft X-ray flux differs because the synthetic baseline uses a constant Gaussian noise floor throughout, while the real reconstruction's quiet-period floor is tightly Kp-constrained, producing a much narrower quiet-time distribution that, when compared against the synthetic version's broader spread, yields a high mean-difference and std.-ratio reading even though both correctly reach storm-level peak values above 10^4 pfu.

WHY THIS APPROACH

This honest, channel-by-channel divergence analysis matters more than a single aggregate 'similarity score' would, because it tells a future engineer exactly where the synthetic dataset is safe to use as a stand-in for real data (testing detection logic, validating model architecture, exercising the decision pipeline) and exactly where it is not (calibrating absolute noise-floor thresholds, tuning fine-grained quiet-period anomaly detectors). Both datasets satisfy the same nine core physics-based assertions identically — Dst < -50 nT, solar wind > 600 km/s, Bz < -20 nT, X8.7-class flares — confirming the synthetic pipeline is a structurally valid approximation of real storm behavior, even where its fine-grained statistical texture differs.

IN PLAIN TERMS

This is the same relationship a flight simulator has to a real aircraft. The simulator gets the big physics right — lift, drag, stall behavior, how the controls respond — well enough to train a pilot meaningfully. It does not perfectly replicate every vibration and engine sound of the real machine. Knowing exactly which parts are faithful and which are simplified is what makes a simulator trustworthy as a training tool rather than a misleading toy, and Table 15.2 is this system's honest account of that same distinction.

16. Discussion

16.1 Strengths

Three strengths distinguish this system from a typical academic forecasting exercise. First, every layer carries its uncertainty forward rather than discarding it: the MC-Dropout flux forecast's standard deviation feeds the upper-confidence-bound used in risk scoring; the MCMC position cloud feeds a probabilistic visibility estimate used in the Stochastic IP; uncertainty is a first-class citizen at every stage, not an afterthought computed once and ignored. Second, the system is explainable by construction, not by a separate interpretability layer bolted on afterward: the 'why' block in Section 14.3 is generated from exactly the same numeric components used to compute the risk score itself, guaranteeing the explanation can never drift from the actual decision logic. Third, and perhaps most distinctively, the thesis documents a genuine engineering failure (the negative-skill forecaster of Section 7.2) and its root-cause diagnosis and fix in full technical detail, rather than presenting only the final, successful configuration — a choice that makes the methodological contribution considerably more useful to a future practitioner than a sanitized success story would be.

16.2 Limitations

This thesis is equally direct about its limitations:

1. **Synthetic-data dominance.** The majority of the forecasting and decision-engine validation runs on engineered synthetic data. While Sections 15.4–15.5 demonstrate strong agreement with the real Gannon storm on the underlying physics, the LSTM forecasting models themselves were not retrained and re-validated directly against the real 44,640-sample Gannon dataset in this iteration of the work — the real-data layer currently validates the physics pipeline (Section 8) but not yet the trained neural forecasters end-to-end.
2. **Orbital model simplifications.** The orbital propagator assumes near-circular, unperturbed two-body Keplerian motion. Real satellites exhibit small eccentricities ($e \approx 0.001$ – 0.003) causing altitude variation of roughly ± 15 km per orbit, and are subject to perturbations — J2 oblateness, atmospheric drag, third-body effects — not modeled here, affecting scheduling accuracy by an estimated 1–2%.
3. **Static priorities and fixed orbital elements.** The ILP formulation uses fixed mission priorities and orbital elements rather than dynamically updated Two-Line Elements (TLEs) from a live tracking feed; a deployed system would need to re-fetch and re-propagate from current TLEs at each scheduling cycle.
4. **Greedy-ILP near-tie is scenario-specific.** Section 13.4's finding that the greedy heuristic ties the ILP's optimal value should not be generalized; it is an artifact of this specific benchmark's low conflict density, and denser ground-station networks or larger fleets would very likely show the greedy heuristic underperforming the ILP by a meaningful margin, as the thesis itself notes.

16.3 Real-World Applicability

The architecture presented here is deployment-shaped, not merely demonstration-shaped: every module — the LSTM forecasters, the physics layer, the orbital propagator, the MCMC sampler, the Markov reliability model, and the ILP scheduler — is already a discrete, independently testable component with a clearly defined input and output contract, exactly the structure a production system

requires. The most direct adaptation path to operational use replaces the synthetic data-ingestion module (Stage 1) with live feeds from actual GOES, DSCOVR/ACE, and Aditya-L1 instruments, and replaces fixed orbital elements with a live TLE feed from Space-Track.org propagated via a standard SGP4/SDP4 propagator — changes that affect only the input boundary of the system, not its internal mathematical structure, which is precisely the point of having built the system this way.

REAL-WORLD ANALOGY

This mirrors how a well-designed power-grid control system is built: the optimization and reliability logic that decides how to route electricity does not need to be rewritten when a utility swaps in a new generation of smart meters — only the data-ingestion layer changes, because the engineers deliberately kept the decision logic decoupled from the specific sensor hardware feeding it. This system was built with the same separation in mind.

17. Future Work

17.1 Real Satellite Integration

The most immediate extension is replacing synthetic orbital elements with live Two-Line Elements fetched from Space-Track.org, propagated through a standard SGP4/SDP4 propagator rather than the simplified two-body Keplerian model used here, and incorporating a genuine RF link budget — free-space path loss, ITU-R P.618 rain fade, and ionospheric scintillation — into the communication-value function of Equation 13.1. A rolling-horizon Model Predictive Control (MPC) scheme, re-solving the ILP over a 2-hour look-ahead window at every 10-minute step as fresh TLE and telemetry data arrives, would convert this from a single-shot scheduling demonstration into a genuinely continuous operational control loop.

17.2 Deep Reinforcement Learning

While the ILP formulation guarantees optimality for any single, fully specified snapshot in time, a Deep Reinforcement Learning (DRL) agent — trained to make sequential scheduling and risk-mitigation decisions directly from the full state (forecasts, orbital geometry, current fleet status) — could in principle learn to anticipate multi-step consequences that a per-snapshot ILP, solved independently at each time step, cannot see: for example, deliberately accepting a slightly suboptimal contact now in order to preserve a satellite's position for a much higher-value contact two time steps later. A natural architecture for this extension is a Graph Neural Network trained on the ILP's own historical optimal solutions (effectively learning to imitate the exact-solver's decisions), providing millisecond-latency near-optimal scheduling suitable for far larger constellations where solving a full ILP at every time step becomes computationally prohibitive.

17.3 Autonomous, Self-Correcting Systems

The long-term vision for this line of work is a fully autonomous digital twin: a live, continuously updated model of the entire constellation and the space-weather environment around it, capable of detecting its own forecast miscalibration (for example, by continuously tracking the 90% confidence-interval coverage statistic introduced in Section 7.5 and triggering an automatic model retraining cycle if observed coverage drifts meaningfully below the nominal 90% target) and re-optimizing its own scheduling and risk-mitigation policy without requiring a human operator to manually notice and intervene. Extending the system's optimization layer to a network-flow formulation incorporating inter-satellite laser or RF crosslinks, and incorporating ITU/IADC space-traffic-management coordination constraints directly into the ILP, would further connect this work to the broader, urgent challenge of managing an increasingly congested orbital environment — the same challenge that motivates the companion Hybrid Satellite-Ground Communication Architecture thesis.

18. Conclusion

This thesis set out to answer a specific, concrete question: given continuously streaming, noisy, and uncertain space-weather telemetry, how should a heterogeneous satellite constellation be monitored, scored for risk, and re-scheduled in near real time, in a way that is both mathematically rigorous and operationally explainable? The system developed and documented here answers that question across five integrated layers — forecasting, physics, geometry, uncertainty, and optimization — each validated on its own terms and then validated together, end to end, against both engineered synthetic scenarios and a real, historically significant geomagnetic superstorm.

Three results stand out as the thesis's central contributions. First, the storm-aware forecasting correction documented in Section 7 transformed a forecaster that was actively worse than doing nothing (skill scores of -10.9 to -52.5) into one that meaningfully outperforms a naive persistence baseline (skill ≈ 0.43 across all horizons) — and the diagnosis of why the original model failed (joint-scaler target leakage, scaled-space clipping, and unweighted loss on a severely imbalanced storm/quiet split) is presented as a transferable lesson for any rare-event time-series forecasting problem, not merely a private debugging note. Second, the per-satellite risk-scoring and decision engine of Section 14, exercised live through three progressively detailed interactive explorers and the companion 3D Three.js scene, demonstrates that a fleet-wide 'all clear' status and individually critical satellites are entirely compatible facts, and that genuinely useful operational decision support requires disaggregated, geometry-aware, explainable scoring at the level of every individual asset, not a single aggregate dial. Third, the validation against the real May 2024 Gannon superstorm, in which all twelve independent physics-based ground-truth assertions passed, demonstrates that the system's underlying equations — not just its machine-learning components, but its physical model of how solar wind energy couples into geomagnetic disturbance — reproduce a real, independently documented historical event with quantitative fidelity.

Taken together with its companion Hybrid Satellite-Ground Communication Architecture thesis, this work demonstrates that the same mathematical toolkit — orbital mechanics, MCMC uncertainty quantification, Markov reliability modeling, and certified-optimal Integer Linear Programming — can be deployed both to make an existing constellation safer and more efficient under space-weather stress, and to question whether so large a constellation needs to exist in orbit at all, when a sufficiently dense ground-based receiver network can substitute for much of its function at a fraction of the cost, the launch risk, and the long-term debris burden. As space traffic continues to grow toward the tens of thousands of active satellites projected by the end of this decade, decision-intelligence systems of exactly this kind — forecasting-aware, uncertainty-honest, and explainable by construction — are not a research luxury. They are the operational infrastructure the next generation of space activity will require.

References

- [1] Vallado, D.A. (2013). Fundamentals of Astrodynamics and Applications, 4th ed. Microcosm Press.
- [2] Battin, R.H. (1999). An Introduction to the Mathematics and Methods of Astrodynamics. AIAA.
- [3] Luenberger, D.G. & Ye, Y. (2016). Linear and Nonlinear Programming, 4th ed. Springer.
- [4] Nemhauser, G.L. & Wolsey, L.A. (1988). Integer and Combinatorial Optimization. Wiley.
- [5] Robert, C.P. & Casella, G. (2004). Monte Carlo Statistical Methods, 2nd ed. Springer.
- [6] Roberts, G.O., Gelman, A. & Gilks, W.R. (1997). Weak convergence and optimal scaling of random walk Metropolis algorithms. *Annals of Applied Probability*, 7(1), 110–120.
- [7] Norris, J.R. (1997). Markov Chains. Cambridge University Press.
- [8] Zhelavskaya, I.S., Shprits, Y.Y. & Spasojevic, M. (2019). Empirical modeling of the plasmasphere dynamics. *Journal of Geophysical Research: Space Physics*.
- [9] Burton, R.K., McPherron, R.L. & Russell, C.T. (1975). An empirical relationship between interplanetary conditions and Dst. *Journal of Geophysical Research*, 80(31), 4204–4214.
- [10] Akasofu, S.-I. (1981). Energy coupling between the solar wind and the magnetosphere. *Space Science Reviews*, 28(2), 121–190.
- [11] Kessler, D.J. & Cour-Palais, B.G. (1978). Collision frequency of artificial satellites: the creation of a debris belt. *Journal of Geophysical Research*, 83(A6), 2637–2646.
- [12] Hochreiter, S. & Schmidhuber, J. (1997). Long Short-Term Memory. *Neural Computation*, 9(8), 1735–1780.
- [13] Bahdanau, D., Cho, K. & Bengio, Y. (2015). Neural Machine Translation by Jointly Learning to Align and Translate. *ICLR*.
- [14] Gal, Y. & Ghahramani, Z. (2016). Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning. *ICML*.
- [15] ITU-R P.618 (2017). Propagation data and prediction methods for Earth-space telecommunication systems. International Telecommunication Union.
- [16] PuLP / COIN-OR CBC Documentation (2023). A Linear and Integer Programming Toolkit for Python. coin-or.github.io.
- [17] GFZ Potsdam (2024). Geomagnetic Kp/Ap Indices, May 2024. German Research Centre for Geosciences.
- [18] NOAA Space Weather Prediction Center (2024). AR13664 Flare and CME Event List, May 2024.

Appendix A: Declaration of Originality and Reproducibility

This thesis documents original computational research. The space-weather decision intelligence system — spanning data ingestion, flare nowcasting, multi-horizon LSTM forecasting with attention and Monte Carlo Dropout, the orbital mechanics and visibility model, the radiation-communication coupling, MCMC orbital uncertainty sampling, the four-state Markov reliability model, the Integer Linear Programming optimizer, the per-satellite risk and decision engine, and the integrated Plotly and Three.js visualization layers — was developed independently in Python (PyTorch, NumPy, SciPy, scikit-learn, PuLP/CBC, Plotly) and in JavaScript/WebGL (Three.js), executed in Google Colab and validated against both an engineered synthetic dataset and a real, GFZ Potsdam-sourced reconstruction of the May 2024 Gannon geomagnetic superstorm.

All numerical results, tables, and figures reproduced in this thesis — including the flare catalog of Table 6.1, the pre- and post-fix forecast metrics of Tables 7.1–7.2, the radiation and communication tables of Sections 8 and 10, the Markov and MCMC diagnostics of Sections 11–12, the ILP assignment and sensitivity results of Section 13, the fourteen-satellite decision snapshots of Section 14, and the twelve-assertion Gannon storm validation of Section 15 — were taken directly from the printed console and figure output of the executed notebook (`radiation_solar_flares.py`, Cells 1 through 17) and the companion `ISRO_SpaceWeather_3D.html` dashboard, without post-hoc alteration of any reported metric.

The companion Hybrid Satellite-Ground Communication Architecture thesis, sharing the same mathematical core, is fully reproducible with a fixed NumPy random seed (42) and the CBC branch-and-bound solver.